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Cosmological parameter inference with Transformer models and simulated Lyman- α forest spectra

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Abstract

We develop a Transformer-based simulation-based inference pipeline capable of predicting cosmological parameters from Lyman- α forest spectra. Because cosmological simulations are often computationally expensive and therefore limited in number, traditional likelihood-based approaches can be difficult to apply, motivating the use of likelihood-free inference methods. The Lyman- α forest is a powerful probe of large scale structure in the universe, and our approach enables direct parameter inference without the need for summary statistics such as the flux power spectrum. We train a Transformer neural network to learn the relationship between spectral absorption features in the Lyman- α forest and the cosmological parameters Ω_m , Ω_b , Ω_Λ as well as the dimensionless Hubble parameter h . To create the training data, we run a suite of 50 cosmological simulations from which we generate the synthetic spectra. The performance of our pipeline is evaluated on both a test set of simulated spectra and on observed spectra from the Sloan Digital Sky Survey (SDSS DR9). We find that the model successfully recovers information about the cosmology from the spectra. In particular, the inferred values of Ω_m and Ω_Λ are consistent with constraints from the Planck 2015 results, while Ω_b shows a systematic underestimation likely related to differences between simulated and observed mean flux levels. The parameter h is the most difficult to infer with the presented setup and our inference mildly favors higher values than those found by Planck. In this work, we demonstrate the potential of Transformer-based approaches for extracting cosmological information from Lyman- α forest spectra while also highlighting the importance of accurate spectral modeling for reliable inference.

Zusammenfassung

In dieser Arbeit entwickeln wir eine auf Transformern basierende Simulation-based-Inference-Pipeline zur Bestimmung kosmologischer Parameter aus Lyman- α -Forest-Spektren. Da kosmologische Simulationen mit hohem Rechenaufwand verbunden sind und daher meist nur in begrenzter Anzahl zur Verfügung stehen, lassen sich klassische likelihood-basierte Methoden oft nur schwer nutzen. Dies motiviert die Nutzung likelihood-freier Inferenzverfahren. Der Lyman- α -Forest stellt eine wichtige Möglichkeit dar, Strukturbildung auf großen Skalen zu beobachten. Unser Ansatz ermöglicht eine direkte Inferenz kosmologischer Parameter aus Lyman- α -Spektren, ohne auf Zusammenfassungsstatistiken wie das Flux-Leistungsspektrum zurückzugreifen. Zur Erstellung der Trainingsdaten führen wir eine Reihe von 50 kosmologischen Simulationen durch, aus denen synthetische Lyman- α -Spektren erzeugt werden. Ein Transformer-Neuronales-Netzwerk wird anschließend darauf trainiert, den Zusammenhang zwischen spektralen Absorptionsmerkmalen und den kosmologischen Parametern Ω_m , Ω_b , Ω_Λ sowie dem dimensionslosen Hubble-Parameter h zu erlernen. Die Leistungsfähigkeit der Pipeline wird sowohl anhand eines simulierten Testdatensatzes als auch anhand beobachteter Spektren aus dem Sloan Digital Sky Survey (SDSS DR9) untersucht. Die Ergebnisse zeigen, dass das Modell kosmologische Information erfolgreich aus den Spektren extrahieren kann. Die inferierten Werte für Ω_m und Ω_Λ stimmen gut mit den Ergebnissen der Planck-2015-Studie überein. Für Ω_b zeigt sich jedoch eine systematische Unterschätzung, die vermutlich auf Unterschiede im durchschnittlichen Flux zwischen simulierten und beobachteten Spektren zurückzuführen ist. Der Parameter h lässt sich mit dem vorliegenden Setup am schwierigsten bestimmen und zeigt eine leichte Präferenz für höhere Werte als in den Planck-Ergebnissen. Diese Arbeit demonstriert das Potenzial Transformer basierter Methoden zur kosmologischen Inferenz aus Lyman- α -Forest-Spektren und unterstreicht zugleich die Bedeutung einer realistischen Modellierung der Spektren für robuste Ergebnisse.

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1 Introduction

1.1 Modern Day Cosmology and Parameter Inference

Cosmology poses questions about the origin, composition, evolution and ultimate fate of our universe. Its aim is to construct a quantitative description of the universe on the largest scales.

The early evolution of the universe can be divided into several phases. Shortly after the Big Bang, the universe was extremely dense and hot. In this primordial state, baryons and their antiparticles were in equilibrium with radiation through constant particle and antiparticle creation and annihilation. After the universe expanded further and thus cooled sufficiently, the radiation lacked the energy to produce new particle-antiparticle pairs. Most of the existing pairs annihilated, leaving behind only a tiny excess of matter due to a not yet fully understood symmetry breaking. The surviving matter is what makes up the entire baryonic content of the universe today. In the following epoch of nucleosynthesis protons and neutrons combined to form the first light nuclei. However, atoms were not able to form until the radiation content of the universe lost even more of its energy to the expansion and decoupled largely from the baryons. This was the beginning of the epoch of recombination, during which electrons combined with nuclei to form neutral atoms. The radiation from this epoch can still be seen today as the Cosmic Microwave Background (CMB).

As this evolution is highly dependent on the expansion of the universe, the description of the rate of expansion sits at the very heart of cosmology. It can be described and understood with a comparatively simple framework and one main equation.

Under the well motivated assumption of a homogeneous and isotropic universe over sufficiently large scales, and by applying Einsteins field equations to a cosmic fluid, Alexander Friedmann (Friedmann 1922; Friedmann 1924) and Georges Lemaître (Lemaître 1927) independently developed a model for an expanding spacetime. This model is expressed by the Friedmann-Lemaître-Robertson-Walker metric and the associated Friedmann equation.

The expansion of the universe is described mathematically with the scale factor $a(t)$. This factor expresses how distances change over cosmic time. It is normalized to be $a = 1$ today, and so a scale factor of $a = 0.5$ would mean, that all distances in the universe were half their distances today. The Friedmann equation connects the Hubble expansion rate $H = \dot{a}/a$ to the energy content of the universe with four energy density parameters,

$$H^2 = H_0^2(\Omega_r a^{-4} + \Omega_m a^{-3} + k a^{-2} + \Omega_\Lambda). \quad (1)$$

H_0 is the Hubble expansion rate today, and each term of the sum describes the contribution of one part of the total energy content of the universe. Radiation is accounted for by Ω_r , matter by Ω_m , curvature by k and dark energy by Ω_Λ . The relative fraction of each value determines not just the expansion rate during the different epochs, but also the global geometry and ultimate fate of the universe. Consequently, measuring these parameters has been a central objective of modern observational astrophysics.

Over the past decade, there has been a number of studies working towards exactly this goal. Observations of the cosmic microwave background performed and analyzed by the Planck Collaboration provide strong constraints on the parameters of the dark energy and cold dark matter (Λ CDM) standard model of cosmology and have established a consistent description of the early universe (Planck Collaboration et al. 2020). For lower redshifts (and thus later times), large-scale structure surveys have mapped the distribution of matter and the expansion history of the universe. The Sloan Digital Sky Survey (SDSS) has played a central role in this effort, as they have measured and collected a large catalog of distant galaxies and quasars, enabling the measurement of the baryonic acoustic oscillations and the analysis of the Lyman- α forest (Lee et al. 2013). More recently, the Dark Energy Spectroscopic Instrument (DESI) has complemented these measurements with extensive observations across a broad range of redshifts (Abdul Karim et al. 2025).

Despite this remarkable observational progress in precision measurements, certain observational discrepancies remain to be solved. One such example is the Hubble tension. Local distance ladder measurements of the present day expansion rate by the SH0ES collaboration (Riess et al. 2022) differ significantly from early-universe observations like the Planck Collaborations’s (Khalife et al. 2024). Such tensions highlight both the sensitivity of recent cosmological observations as well as the importance of the development of robust inference methods capable of extracting information from increasingly complex and large data sets.

1.2 Large-Scale Structure and the Intergalactic Medium

A key probe of cosmology is to study the large-scale structure of the universe. It arises from the gravitational growth of primordial density fluctuations in the early universe. Small perturbations in the matter distribution evolve over cosmic time and under the influence of gravitational instability they form a hierarchical network of structures that include filaments, sheets and halos. Regions with slightly higher density attract additional matter, gradually growing into the structure we see today. This process directly links cosmological parameters to the statistical distribution of matter on large scales.

While dark matter (DM) dominates the gravitational behavior and thus the structure formation, baryonic matter falls into its gravitational potentials, tracing these structures and allowing for indirect observation. On large scales most baryonic matter ($\approx 90\%$) resides in the Intergalactic Medium (IGM) (Shull et al. 2012), the gas filling the space between galaxies. The state of the IGM is determined by an interplay of multiple different physical mechanisms, such as cosmic expansion, gravitational collapse, shock heating from large-scale structure formation, energy input from quasars, hot stars and supernovae contributing to the UV background (UVB) radiation (Tejos 2025). As a result, the distribution of baryonic matter does not simply mirror the DM distribution, but rather reflects a complex interplay of different physical mechanisms over a wide range of scales. Nonetheless, the state of the IGM is sensitive to changes in cosmological conditions, making it a great tool for probing variations in the expansion rate, matter content and amplitude of density fluctuations. Consequently, observables that probe the IGM offer a means of connecting cosmological parameters to measurable quantities along lines of sight through the universe.

The main observational method to measure the IGM depends on its interaction with background radiation. As light travels through cosmic structures, it passes through regions of varying density that imprint characteristic absorption signatures. These absorption features encode information about the distribution and physical state of matter.

1.3 Lyman- α Forest as Cosmological Probe

A particularly powerful observable of the intergalactic medium is the Lyman- α forest, a series of absorption features observed in the spectra of distant quasars. As photons emitted by a background quasar propagate through the universe, they encounter regions of neutral hydrogen that absorb radiation at the Lyman- α transition. Because the photons travel over cosmological distances, the expansion of the universe continuously redshifts their wavelengths. As a result, photons that were originally emitted at shorter wavelengths are shifted into the range of the Lyman- α transition at different points along the line of sight. The observed spectrum therefore exhibits a forest-like dense sequence of absorption features blueward of the intrinsic Lyman- α emission line of the quasar. For an example and visual explanation see Fig. 1.

The strength and distribution of these absorption features depend on the density and physical state of the intervening hydrogen gas. Consequently, the statistical properties of the Lyman- α forest encode information about cosmic structure across a wide range of redshifts, otherwise difficult to access observationally.

The connection between Lyman- α absorption and cosmological evolution is deeply rooted in the ionization

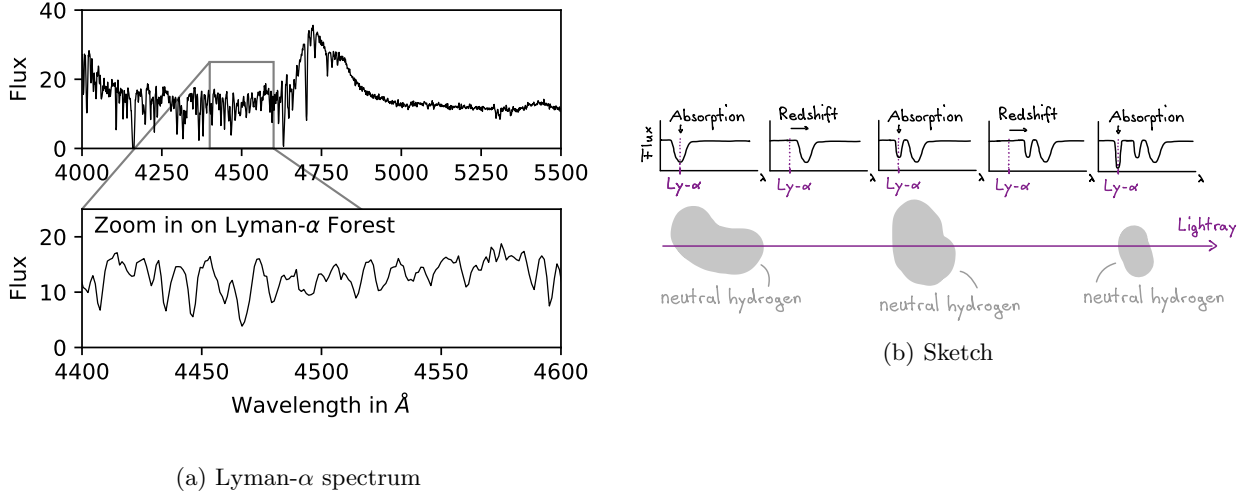


Figure 1: Example of a Lyman- α forest spectrum and a sketch of the mechanism creating it. In (a) an example of a quasar spectrum from the SDSS Catalog (Lee et al. 2013) with a zoom in on a part of the Lyman- α forest region can be seen. In (b) a sketch of the mechanism that creates Lyman- α forest spectra can be seen. Neutral hydrogen at different redshifts absorbs the Lyman- α line, leading to the forest-like absorption features.

history of the IGM. Following recombination (the epoch where the first atoms formed), the baryonic content of the universe consisted predominantly of neutral hydrogen. As the first generations of stars and quasars formed, their radiation progressively ionized the surrounding gas in a process known as cosmic reionization. This transition did not occur uniformly throughout space, instead, ionized regions expanded and eventually overlapped, producing a patchwork of ionized and neutral environments whose properties depend on both cosmological parameters and astrophysical processes.

At redshifts below approximately $z \sim 6$, most of the intergalactic medium is highly ionized but retains a small residual neutral hydrogen fraction. It is this diffuse, partially ionized gas that gives rise to the Lyman- α forest observed in quasar spectra over the approximate range $6 \gtrsim z \gtrsim 2$. The absorption signal depends sensitively on the density distribution of matter as well as the thermal and ionization state of the gas. Therefore, interpreting Lyman- α spectra requires a detailed physical model of structure formation and the evolution of the intergalactic medium. The complexity of these coupled processes makes accurate theoretical predictions challenging and motivates the use of numerical simulations.

1.4 Numerical Simulations of the Lyman- α Forest

Over the past decades, numerous cosmological simulation projects have been developed, often with differing scientific priorities. Some focus primarily on the evolution of large-scale structure across cosmological volumes such as Eagle (Schaye et al. 2014) or IllustrisTNG (Nelson et al. 2021), while others aim to resolve the detailed physics of individual galaxies or even smaller-scale processes, as pursued in projects like Auriga (Grand et al. 2017) or Nihao (Wang et al. 2015). As discussed in the previous sections, the Lyman- α forest probes matter distribution on cosmological scales. Consequently, the simulations employed in this work follow the former approach, modelling large cosmological volumes while incorporating essential baryonic physics. At the same time, physical processes occurring on subgalactic scales cannot be neglected, as they influence the thermal and ionization state of the intergalactic medium.

In this work, we run simulations with the TNG-AREPO code (Pillepich et al. 2017), which evolves dark matter and baryonic gas by solving gravity and magnetohydrodynamics equations on a moving mesh. The

underlying numerical scheme models a broad range of astrophysical processes relevant for galaxy formation, including star formation, stellar evolution, chemical enrichment, primordial and metal-line gas cooling, stellar feedback driven galactic outflows, and supermassive blackhole formation, growth, and feedback (Pillepich et al. 2017). Because key processes such as star formation and black hole accretion occur on spatial (and mass) scales below the resolution of cosmological simulations, they are implemented through subgrid models that capture their effective large-scale impact. The TNG-AREPO framework is an extension to the Arepo code (Vogelsberger et al. 2013) and introduces several refinements compared to earlier implementations, including the addition of magnetic fields, as well as improved modeling of the growth and feedback of supermassive black holes, galactic winds, and the stellar evolution and gas chemical enrichment (Pillepich et al. 2017).

A crucial aspect for modelling the Lyman- α forest is the ionization state of hydrogen in the intergalactic medium. The dominant mechanism governing this state is radiative transfer of ionizing photons, which is computationally prohibitive to model self-consistently in large cosmological volumes (Kannan et al. 2019). Instead of simulating it directly, a spatially uniform, time-dependent ultraviolet background (UVB) radiation field with substantial contributions from quasars and star-forming galaxies at $z < 3$ is assumed, following Faucher-Giguère et al. (2009). This model has specifically been shown to be in good agreement with the measured Lyman- α forest opacity at redshifts $z = 2 - 4.2$, and to be compatible with hydrogen reionization completing by $z = 6$ (Faucher-Giguère et al. 2008). This UVB heats and ionizes the gas and thus determines how much neutral hydrogen remains after reionization. However, this background radiation should be weaker in dense regions of neutral hydrogen, so a self-shielding mechanism is implemented, where the local gas attenuates the incident radiation and reduces the effective ionization rate in dense regions.

Together, these modelling choices enable the simulation to capture the dominant physical mechanisms that shape the intergalactic medium on cosmological scales while maintaining computational feasibility. At the same time, the reliance on an assumed ultraviolet background and subgrid prescriptions introduce systematic modelling assumptions that must be considered when interpreting simulated Lyman- α spectra and comparing them to observations.

1.5 The Likelihood Problem and Simulation-Based Inference

The simulations described above provide a forward model for Lyman- α spectra for a given set of cosmological parameters. The central objective of this work, however, is to perform inference in the opposite direction, constraining the parameter space from observed spectra. In principle, we could now proceed to use standard Bayesian inference methods such as Markov chain Monte Carlo (MCMC) (Gilks et al. 1995) or variational inference (Zhang et al. 2019). These methods require evaluating the likelihood of the observed data given a set of parameters.

In the context of our simulations, constructing such a likelihood function analytically is unfeasible, as the mapping from cosmological parameters to our observable spectra includes nonlinear gravitational evolution, hydrodynamics, thermal processes and subgrid modeling of complex astrophysics. Furthermore, this mapping is only accessible through computationally expensive simulations. We could construct approximate likelihood functions using summary statistics like the power spectrum or the probability density function (PDF) of transmission, but would lose great amounts of information, as these summary statistics are not optimized for this task (Nayak, Parth et al. 2024). These limitations motivate us to use inference methods that can directly constrain parameters from simulated data without the need for handcrafted explicit likelihood functions.

Simulation-based Inference (SBI) is a powerful set of tools for exactly this purpose. The key idea behind SBI is to generate a training set by drawing parameters θ from a prior distribution and simulating the corresponding data x . An inference network (usually a neural network) is then trained on x to learn the relationship between the simulated data and parameters. Once trained, the model can be used to perform inference on observed data x_0 . By relying only on model simulations, SBI methods do not require explicit evaluations of likelihoods or gradients, making them applicable to any simulator (Deistler et al. 2025). Another key advantage of this

approach is amortization: the computationally expensive simulations and training of the inference network is performed only once, after which inference can be carried out on new observations rapidly and without the need for retraining.

Mathematically our network can be seen as a deterministic estimator $f(x_i) = \hat{\theta}_i$ that maps from the data (our spectra) x_i to predicted parameters $\hat{\theta}_i$. The simulation of Lyman- α forest spectra can be seen as sampling from the probability distribution $x_i \sim p(x | \theta_*)$, as we generate many different spectra for one set of true parameters θ_* . Thus, one point prediction of our network follows the distribution

$$\int p(\hat{\theta} | x) p(x | \theta_*) dx = \int \delta(\hat{\theta}_x - f(x)) p(x | \theta_*) dx = p(\hat{\theta} | \theta_*), \quad (2)$$

where we can write $p(\hat{\theta} | x) = \delta(\hat{\theta}_x - f(x))$, as the network is deterministic and so the probability distribution over $\hat{\theta}$ collapses to a delta function given a specific x . The distribution $p(\hat{\theta} | \theta_*)$ in itself is not a posterior distribution, but rather a sample distribution in a frequentists setting. It tells us how the pipeline performs with repeated inference given a ground truth. This already is very useful to quantify properties like biases and variance of the pipeline. Our goal, however, is to make a statement about the true parameter θ_* . For this purpose we want to construct a posterior from $p(\hat{\theta} | \theta_*)$. This can be achieved, by applying our inference network to many different spectra (all drawn from the same ground truth θ_*), sampling the likelihood function

$$p(\hat{\theta}_1, \dots, \hat{\theta}_n | \theta_*) = \prod_{i=1}^n p(\hat{\theta}_i | \theta_*). \quad (3)$$

Assuming a prior $p(\theta_*)$ (in our case this prior is assumed to be uniform over the entire parameter space), we can use Bayes Theorem to obtain the posterior distribution

$$p(\theta_* | \{\hat{\theta}_i\}) \propto p(\theta_*) \prod_{i=1}^n p(\hat{\theta}_i | \theta_*), \quad (4)$$

where $\{\hat{\theta}_i\}$ can be interpreted as a summary statistic of the data $\{x_i\}$. Thus, the inference network effectively learns a custom summary statistic that preserves as much Fisher information about θ_* under the simulator as possible¹.

1.6 Choice of Model Architecture and Inductive Bias

Choosing a specific neural network architecture for the inference model inherently introduces an inductive bias. In machine learning, an inductive bias is understood to be the set of assumptions a model uses to generalize from training data to previously unseen inputs. These assumptions are encoded in the model architecture, training procedure, and parameterization, and they determine which relationships are easier for the model to learn (Shalev-Shwartz and Ben-David 2014). Choosing a model architecture and thus an inductive bias that benefits the type of data it is applied to is central. This ensures both model efficiency and accuracy of inference.

Different neural network architectures impose quantitatively different biases. Multi-layer perceptrons, for example, generally assume only the smoothness of the approximated function, and thus introduce a relatively weak inductive biases. This makes them applicable to a broad variety of problems, but also limits their scalability and data efficiency when applied to high-dimensional data. In contrast, architectures that explicitly benefit from known structure in the data can achieve improved performance by embedding domain knowledge directly into the model. This is especially useful in physics, as the data-generation is typically governed by well-understood physical processes.

¹A similar idea and even deeper mathematical intuition can be build in a measure theory setting (see Appendix 6.2).

Spectra have some inherent properties, that make a Transformer architecture particularly well suited for inference. First, they are naturally represented as an ordered sequence of single wavelength measurements. This sequential structure allows Transformers to treat spectra as a set of tokens, similar to natural language, which this architecture was originally designed for (Vaswani et al. 2023). Secondly, the physical processes responsible for creating the spectra can produce correlations between widely separated wavelengths. For example, large-scale density fluctuations can imprint patterns that are not strictly local in wavelength space. Transformer models are specifically designed to capture such long-range dependencies through the self-attention mechanism (Vaswani et al. 2023). Self-attention allows the model to relate the importance of each wavelength to every other wavelength directly, making it possible to learn global relationships in the spectra. This distinguishes Transformers from other architectures such as Convolutional Neural Networks (CNNs), who rely on fixed kernel sizes, inducing an inductive bias towards local relationships (Lecun et al. 1998). This motivates our choice of using a Transformer model as the inference network.

1.7 Transformer Models

The Transformer architecture was first developed by Vaswani et al. (2023), and has been hugely successful since, being applied in a variety of fields, such as Natural Language Processing (Patwardhan et al. 2023), Medical Image Segmentation (Chen et al. 2021), Bioinformatics (Ghosh et al. 2025) and time-series forecasting across diverse domains (Wang et al. 2024). As explained in the previous section, this architecture is a natural choice for the problem of parameter inference with spectra, and so we build a dedicated Transformer model for this purpose, following closely the original design.

The main building blocks of a Transformer model can be seen in Fig. 2. Every input token (in our case wavelengths) gets embedded into a higher dimensional space. The Transformer layers then learn a meaningful representation of the embedded input and its patterns in this space. This is then given to a final MLP head to make the prediction in the parameter space.

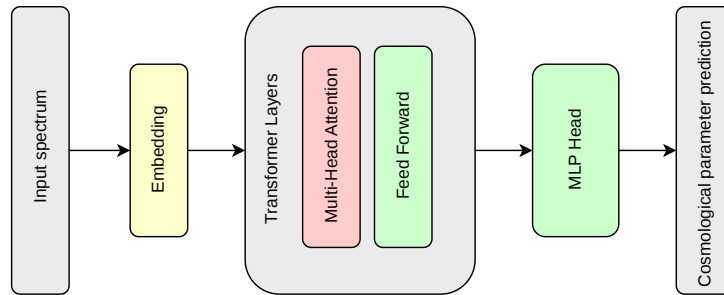


Figure 2: Flow chart of the most basic building block for a general Transformer model.

Each Transformer layer consists of mainly two important blocks. The first one being a Multi-Head Attention block. The main idea, is to let the model learn the importance of each input token for every other input token. Mathematically, every attention head projects the input into three vectors: a query Q , a key K and a value V . The attention score is then calculated by comparing queries and keys,

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_k}} \right) V, \quad (5)$$

so that the query represents a request for the relevant context between tokens, the keys encode which tokens carry the relevant information, and the value provides the content that is combined to form the output. d_k is the feature dimension of each head, and the factor of $1/\sqrt{d_k}$ is mainly there to stabilize training and prevent the dot product from becoming too large as the dimensionality increases.

The Multi-Head Attention block lets the network learn the relation between different tokens. The outputs then go through the second main block: a feed forward network, that projects them into a higher dimensional space, and then back into the original feature space. This feed forward network learns how to represent the tokens in a meaningful way with the updated information of the Attention Heads. It is this combination of learning the relationships between tokens and then updating and learning the representations of each token individually that makes Transformers so powerful.

1.8 Scope and Contributions

The goal of this thesis is to investigate simulation-based inference with spectral data. More specifically, this work explores whether the neural network architecture of Transformer models can efficiently learn the relationship between absorption patterns in the Lyman- α forest and the underlying cosmological parameters using simulated training data.

To address this problem, we forward model Lyman- α forest spectra for different cosmologies using the magnetohydrodynamic TNG-Arepo code. We then use these synthetic spectra to train a Transformer model and evaluate its performance. The trained model is then applied to real spectra observed by the SDSS survey to assess its robustness and applicability.

The scope of this work is limited by many systematic uncertainties and approximations. We focus on the methodological question of how Transformer models can be used in a simulation-based inference setting to extract information about the cosmology given spectral data. This work should not be seen as an attempt to derive new constraints on the cosmological parameters, but rather as a methodological proof-of-concept.

The following sections are structured as follows: In Section 2 we explain the simulation setup in detail, describe the initial conditions and choice of parameters, as well as the sampling of the parameter space and the creation of the spectra. In Section 3 we provide insight on the machine learning methods, including data preprocessing and the creation of our datasets and training procedures. In Section 4 we present the results of our parameter inference on test spectra and observed spectra as well as the model performance, and finally in Section 5 we discuss our findings and give future directions.

2 Simulation Methods

To obtain training data for the Machine Learning models, we run a suite of cosmological simulations (called CLIMB - **C**osmological **L**ambda **C**DM Simulations for **I**nference with **M**achine Learning and **B**ayesian Statistics) using the TNG-AREPO code. TNG-AREPO is a moving mesh magnetohydrodynamics + gravity code, that has been used in large simulation projects such as IllustrisTNG (Nelson et al. 2021) or TNG-Cluster (Nelson et al. 2024). We choose to vary four different cosmological parameters (Ω_m , Ω_b , Ω_Λ , H_0) in our suite, sampled using a Latin Hypercube. To create our initial conditions, we use the MUSIC code (Hahn and Abel 2011), and keep the random initial seed fixed for our higher resolution run. We make 10,000 spectra for each box using the TEMET package (Nelson et al. 2025), that traces rays through the Voronoi mesh and calculates the relevant absorptions. The spectra are created for snapshots of the simulations at $z = 2$. Finally, we augment the spectra to increase their length, by patching randomly shuffled spectra from the same box together. Using this pipeline, we create 500,000 spectra in total that we can use to train the inference models. Additionally, we run a reference box, with cosmological parameters consistent with Planck measurements ($\Omega_\Lambda = 0.6911$, $\Omega_m = 0.3089$, $\Omega_b = 0.0486$, and $h = 0.6774$) (Planck Collaboration et al. 2016), to serve as a benchmark test of the inference model.

We run one higher resolution simulation suite (256^3 particles) called CLIMB high and two lower resolution suites (128^3 particles) called CLIMB low, each with a different strategy to choose the initial conditions, as we also want to study the impact of the initial conditions on the inference quality.

2.1 Choice of Simulation Parameters and Simulation Setup

We now outline the choice and ranges of the cosmological parameters Ω_m , Ω_b , Ω_Λ , and H_0 , varied in the CLIMB suits. In our inference pipeline, we in fact use the dimensionless Hubble parameter h instead of H_0 , which is defined as $H_0 = 100h$ km/s/Mpc. One might realize, that we vary all of the parameters from the Friedmann Equation (see Eq. 1) except for the density parameter of radiation Ω_r and the curvature parameter k . We set both of them to zero for all simulated boxes. The energy density of the universe is heavily dominated by Ω_m for redshifts lower than $z \approx 3700$ in Λ CMD until, for high redshifts, Ω_Λ becomes increasingly important (for a derivation of this, see Appendix 6.1). As our simulations start at $z = 127$, $\Omega_r = 0$ is a very good approximation. Setting $k = 0$ is a standard choice for such simulation suites, as multiple independent studies consistently find a flat universe (Dias et al. 2025; Vagnozzi et al. 2021). Also, this assumption is motivated theoretically by cosmic inflation theory, and it is not clear how a curved spacetime would affect the Lyman- α forest spectra or simulations. We thus leave this for future studies to explore.

The chosen parameters and their ranges for this work can be seen in Table 1. For the ranges, we choose relatively wide intervals as we want to be confident that there is enough variability in the spectra for the inference network to learn the differences in cosmologies. One could, of course, also choose tight measurement constraints as interval ranges and thereby try to derive new and stronger constraints on the cosmological parameters. As we have discussed in Sec. 1.8, however, this is not the goal of this work, and so this also is left for future studies to explore.

Table 1: Parameters of the CLIMB simulation suite.

Constant parameters	Value		Varied parameters	Range
Box size	25^3 Mpc ³		Ω_m	[0.1, 0.5]
Particles per box (CLIMB high)	256^3		Ω_b	[0.001, 0.1]
Particles per box (CLIMB low)	128^3		Ω_Λ	[0.5, 0.9]
# of Sims (CLIMB high/CLIMB low)	50 / 2×50		h	[0.55, 0.85]

A second set of important parameters for such simulation suits is the box size, resolution and number of simulated boxes. We run scaling tests for both the box size and number of particles per box. In our main suite (CLIMB high) we choose to run 50 simulations (+1 reference box) with a box size of 25^3 Mpc^3 and 256^3 particles for both dark matter and gas particles. A comparison to other existing simulation suits with similar goals can be seen in Fig 3. It is worth mentioning, that both CAMELS and DREAMS have very similar box sizes and Baryon Mass Resolutions to the CLIMB suits. However, the DREAMS suite was mainly made to probe alternate DM models (Rose et al. 2024), and the CAMELS suite does not only vary cosmological parameters, but also astrophysical parameters (Villaescusa-Navarro et al. 2021), and thus both of these do not contain a set of simulations fit exactly for our purpose.

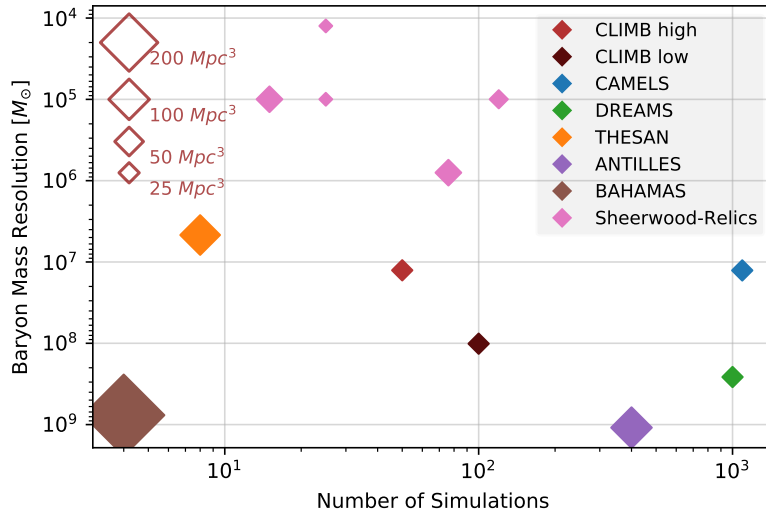


Figure 3: Overview of existing simulation suits with varying cosmologies. These include CAMELS (Villaescusa-Navarro et al. 2021), DREAMS (Rose et al. 2024), THESAN (Kannan et al. 2022), ANTILLES (Salcido et al. 2023), BAHAMAS (McCarthy et al. 2016) and Sheerwood-Relics (Puchwein et al. 2022).

2.2 Initial Conditions

Cosmic structure formation is driven by the gravitational growth of primordial density fluctuations. Consequently, constructing accurate initial conditions (ICs) is crucial for any cosmological simulation. To archive this, we use the MUSIC code (Hahn and Abel 2011) for $z = 127$. MUSIC creates ICs for both baryonic and dark matter. It computes real space transfer functions and combines them with white noise random fields, to compute a seed density field. Initial displacement and velocity fields, based on solutions of Poisson's equation, are calculated following first and second order Lagrangian perturbation theory. The density and velocity transfer functions are calculated for the baryonic matter and dark matter separately, as baryon density fluctuations are not yet exactly tracing the dark matter fluctuations at high redshifts.

Our simulated box sizes might not be large enough for all initial seeds to produce the same statistical distributions. To combat this, we keep the initial random seed fixed for all boxes of our high resolution suite. We thus make sure, that the differences in the boxes are purely a product of the differences in cosmology, and do not depend on the initial random seeds. This, however, might introduce a bias itself, as we can not be sure, that the chosen seed is perfectly representative of the real universe. To study the impact of changing the initial random seed, we run two lower resolution suites, one with a constant initial random seed (as reference), and one with a different initial random seed for every box. We can thus make a comparison between methods without the need to rerun the full (and computationally expensive) high resolution suite.

2.3 Latin Hypercube Sampling

To choose the actual points in parameter space that we run simulations for, we use Latin Hypercube Sampling (LHS). A Latin Hypercube is constructed by dividing each dimension into equally spaced cells. Within each cell, one sample point is placed randomly. The sample points of the different dimensions are then randomly shuffled against each other. This creates a sampling that is less prone to clustering, especially for a low number of samples in higher dimensional parameter spaces (McKay et al. 1979). To further prevent clustering and improve sampling of the entire space, we also implement a Monte Carlo approach, following Bird et al. (2019). We simulate 10,000 Latin Hypercubes, and use the one with a maximized space filling metric. Our space filling metric is the sum of Euclidean distances of the nearest neighbor of each point.

As the cosmological parameters are not fully independent of each other, we obtain two conditions that constrain our four hypercube dimensions further, with

$$\Omega_m \geq \Omega_b, \quad (6)$$

$$\Omega_m + \Omega_\Lambda = 1 \quad (7)$$

The first condition (Eq. 6) is simply a consequence of the definition $\Omega_m = \Omega_c + \Omega_b$, where Ω_c is the density parameter for dark matter. The second condition (Eq. 7) follows from the fact, that we have set $k = 0$ and $\Omega_r = 0$ in this work, as explained in Section 2.1. Thus, Ω_m and Ω_Λ are the only two components left in the universe, and so naturally they need to sum to one, giving us a full degeneracy between them. As a consequence, we only need to sample a 3-dimensional Latin Hypercube and can simply calculate the fourth parameter. The condition in Eq. 6 proves to be less straightforward to enforce, as one usually assumes the sampled parameters to be completely independent for LHS. As a simple workaround, we choose to sample Ω_m and Ω_b independently, constructing the full hypercube and then check the condition for each point individually, removing it, if it is not met. To still obtain the desired amount of sampling points, we construct a second hypercube (again checking the condition), and complement the first one with the needed amount of sample points. We repeat this process if needed.

We choose LHS over sampling the parameter space with a uniform grid, as uniform grids scale poorly with higher dimensional spaces. In our 3-dimensional cube, we could only place around $\sqrt[3]{50} \approx 3.7$ grid points per dimension, which makes a very poor sampling. A 2-D visualization of a comparison between LHS, random sampling and a uniform grid can be seen in Fig. 4.

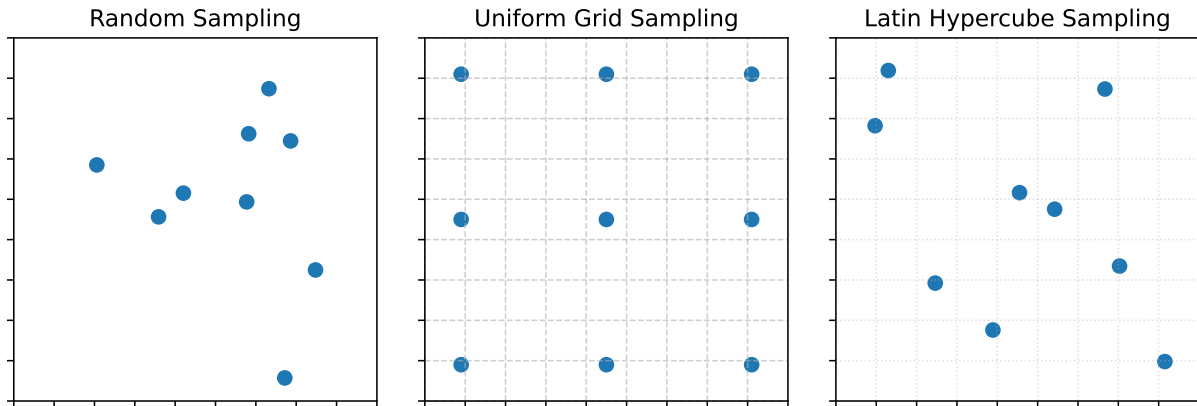


Figure 4: Comparison of different sampling methods for 2D box. On the left: Random sampling is prone to under sampling regions (lower left of the box), in the middle: Uniform Grid gets exponentially more expensive with each dimension, on the right: LHS covers most regions of the box.

2.4 Sanity Checks of the Simulations

Before generating spectra and using the simulations for training the inference models, it is useful to verify that the simulated boxes reproduce the expected large-scale structure and basic properties of galaxy formation. As a first sanity check, we visualize projections of the neutral hydrogen density for a subset of simulation boxes. Fig. 5 shows projections for 20 boxes from the CLIMB-high suite at redshift $z = 2$, ordered by increasing baryon density parameter Ω_b . The characteristic filamentary structure of the cosmic web is clearly visible, with dense nodes connected by extended filaments and surrounded by large underdense regions. These structures arise from gravitational collapse in the underlying dark matter distribution and represent the large-scale environment in which the Lyman- α absorption features originate. Differences between the cosmological models are visible in the amount and distribution of neutral hydrogen, reflecting the varying baryon content in the simulations.

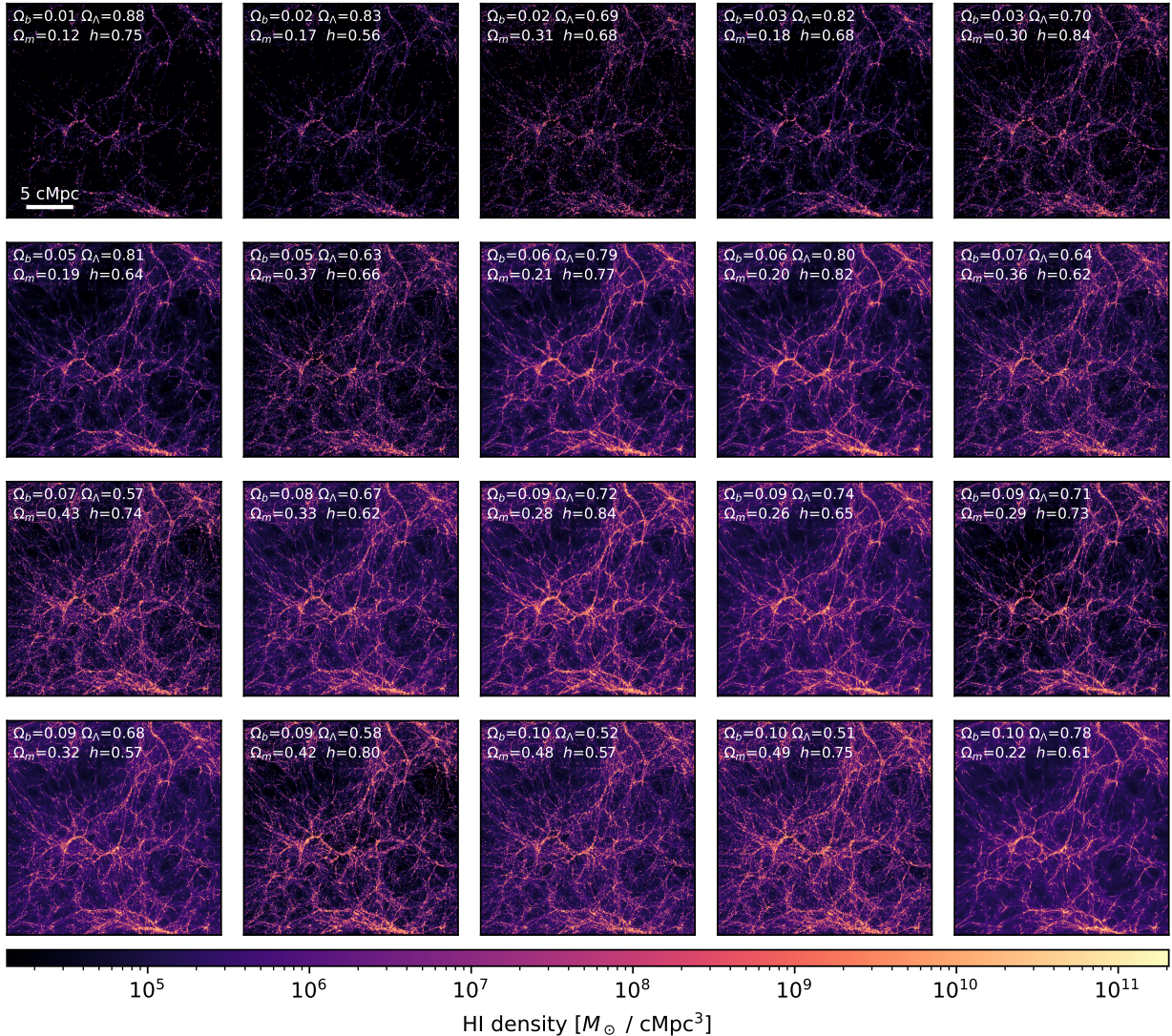


Figure 5: Projection of the mean density of neutral hydrogen at $z = 2$ for 20 of the CLIMB high boxes ordered by Ω_b . The filamentary cosmic web structure is clearly visible, with variations in the neutral hydrogen distribution reflecting differences in baryon content between the cosmologies.

Zooming in on individual halos provides further insight into the internal structure of the simulated galaxies. Fig. 6 shows the distributions of stars, gas, and dark matter for the most massive halo in ten randomly selected simulation boxes at $z = 0$. As expected, the majority of the mass is concentrated near the halo center, with the stellar component forming a compact central structure embedded within a more extended gas distribution. Both the stellar and gas components trace the underlying dark matter halo, illustrating the coupling between baryonic and dark matter dynamics in the simulations. While the detailed morphology varies between boxes due to differences in cosmological parameters and stochastic structure formation, the overall structure of the halos is consistent with expectations from galaxy formation models.

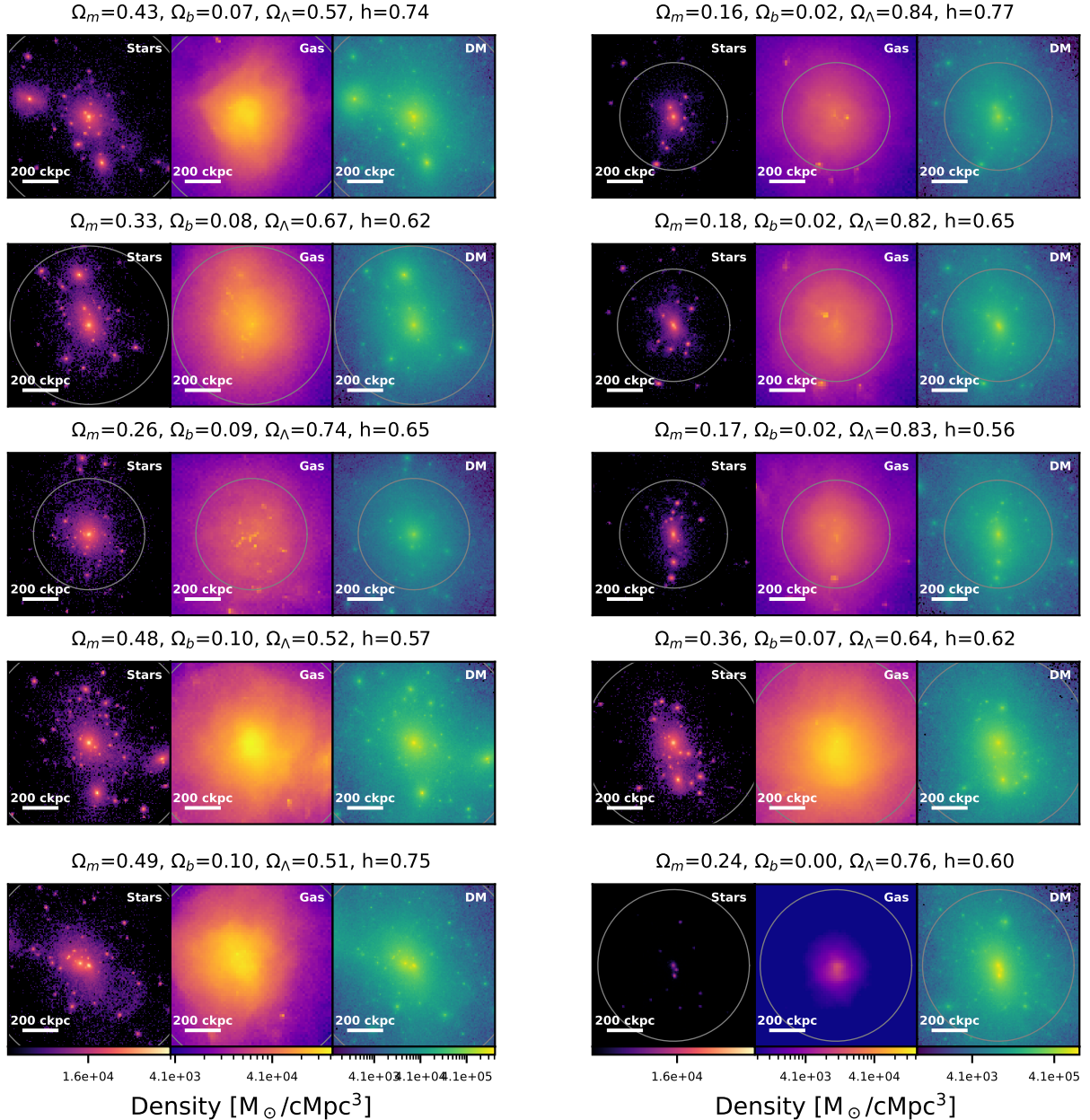


Figure 6: Stars, Gas and Dark Matter for the most massive halo in 10 of the boxes from the CLIMB high suite at $z = 0$. The gray circles represent the regions, where the mean density is larger than 500 times the critical density of the universe.

A complementary summary statistic is provided by the halo mass function, which describes the abundance of halos as a function of their mass. The halo mass functions for all 50 simulation boxes are shown in Fig. 7. Overall, the shape of the halo mass function in each box follows the expected form and agrees well with reference curves from the dark-matter-only runs of the IllustrisTNG simulations (Nelson et al. 2021). A systematic shift in the overall normalization of the halo mass function can be seen as a function of the matter density parameter Ω_m , with lower values of Ω_m producing fewer halos across all mass bins. This behavior is expected, as a lower matter density reduces the amount of gravitational clustering and therefore the abundance of collapsed structures. The mass functions also exhibit a cutoff at the low-mass end due to the finite resolution of the simulations, as well as a cutoff at the high-mass end caused by the limited simulation volume.

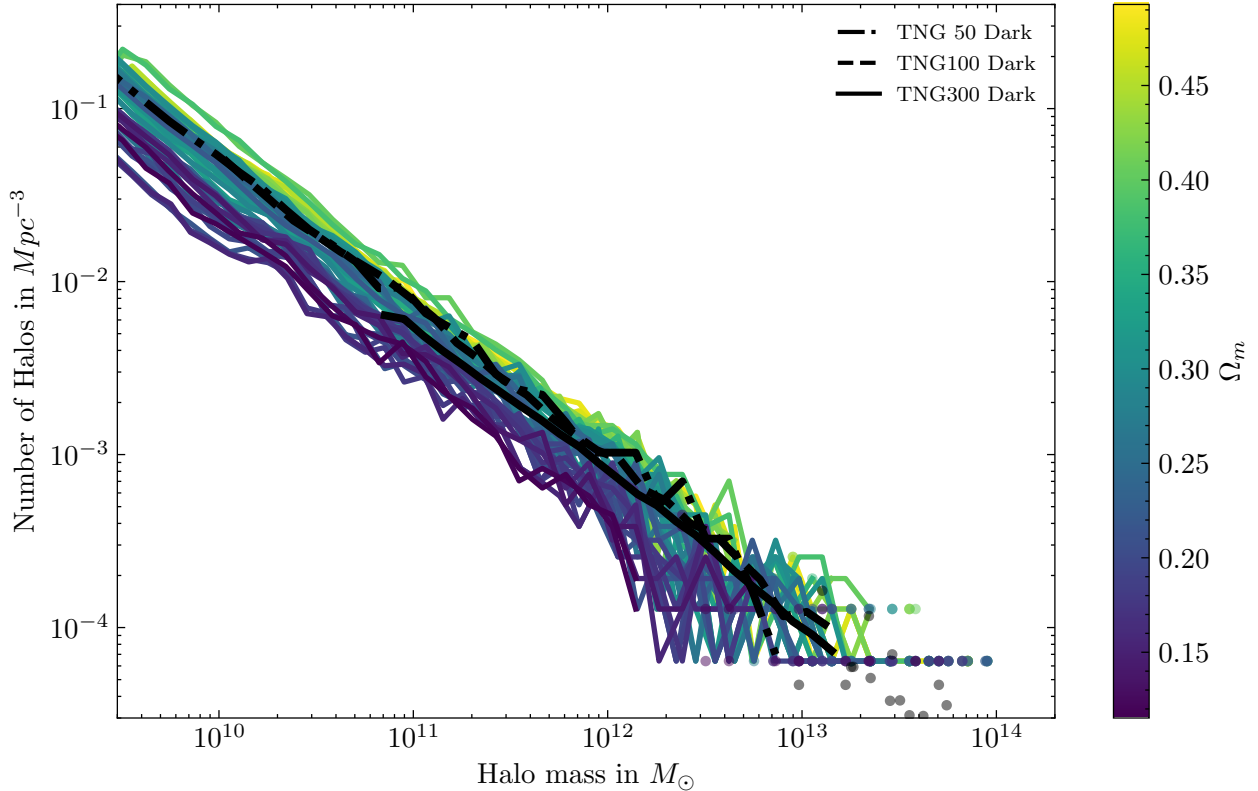


Figure 7: Halo mass function for all 50 simulated boxes of the CLIMB high suite. The shape of all lines have a similar slope, however they show a shift in the number of halos for different cosmologies. Three reference lines from the TNG50, TNG100 and TNG300 simulations (Nelson et al. 2021) are shown as black lines.

Taken together, these visual and statistical checks confirm that the simulations produce physically plausible large-scale structure and halo populations across the explored cosmological parameter space. Having established the validity of the simulated environments, we can now proceed to generate synthetic Lyman- α forest spectra from the simulation outputs.

2.5 Absorption Spectra

We save snapshots of the simulations at redshift $z = 2$, to make the Lyman- α spectra in the following post-processing steps. To generate the spectra we use the TEMET package (Nelson et al. 2025). It uses a novel ray tracing algorithm through the Voronoi mesh to create an ordered list of intersections of Voronoi

cells and their path-lengths to then compute the continuum-normalized absorption spectrum for each sight line. This is done on a high resolution wavelength grid, where the absorption is computed by loading the physical properties of the gas cells (temperature, line of sight velocity and density of the particular ion) and calculating the corresponding Voigt absorption profile for the optical depth. TEMET supports specific instrument configurations such as the BOSS spectrograph, and so the spectrum is convoluted with the instrument line spread function (LSF) and finally, resampled onto the instrumental wavelength grid. This leaves us with idealized noise-free synthetic spectra modeled for the BOSS spectrograph. We use TEMET to make 10,000 spectra for each box at redshift $z = 2$ along a uniform 100 by 100 grid of sight lines that traverse the box in parallel. Examples of these spectra can be seen in Fig. 8

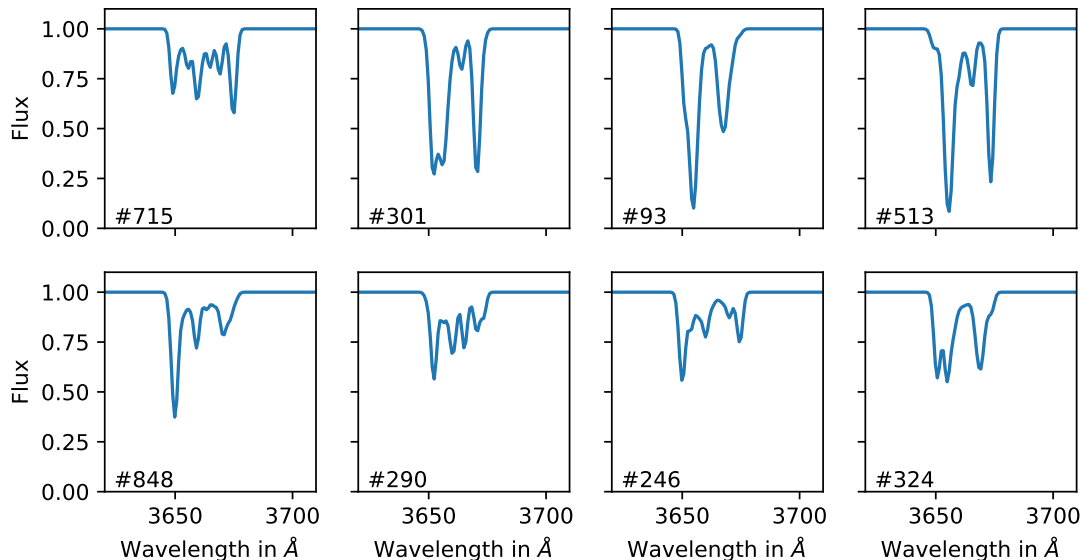


Figure 8: Example gallery of single spectra created from the CLIMB high simulations using TEMET (Nelson et al. 2025). All spectra shown were made from the same box and different lines of sight.

To test whether the different cosmologies imprint themselves into the spectra, we plot them for visual confirmation. We plot the spectra from the same lines of sight for every box in Fig. 9, where we can see a clear dependence of the depth and width of absorption features on Ω_b . This is exactly what we would expect, as the baryon density dictates the amount of absorbing gas in the universe. In Fig. 10a we stack spectra from the same lines of sight for every box in a 2D plot and sort for Ω_Λ . The most noticeable difference is that the width of the spectra depend on cosmology. This is a fully expected effect, as the scale factor will grow at different rates with time for different cosmologies. Thus, the different boxes will have varying physical box sizes (they are all 25^3 Mpc^3 in comoving distance, but certainly not in physical distance). So despite all snapshots being taken at $z = 2$, the age of the universes will vary depending on cosmology (see Fig. 10b). Even though expected, we have to be careful about the way we treat this, as the absorption features being closer together is a physical effect encoding Ω_m , Ω_Λ and h , but the different widths of the spectra is an artifact of the limited box sizes in the simulations.

To combat the issue that the spectra width is a result of limited box size in simulations, we patch multiple spectra together, which also has another major advantage. We can make the spectra significantly longer, which means each spectrum will carry more information, making every point estimate of our inference network more informed. In training this will not make a relevant difference, as the total amount of information in the training set does not increase. However, we can use longer sections of observed spectra in the inference process, which does increase the amount of information the network has access to.

To patch spectra together, we first have to calculate the width of the spectra from the cosmological parameters.

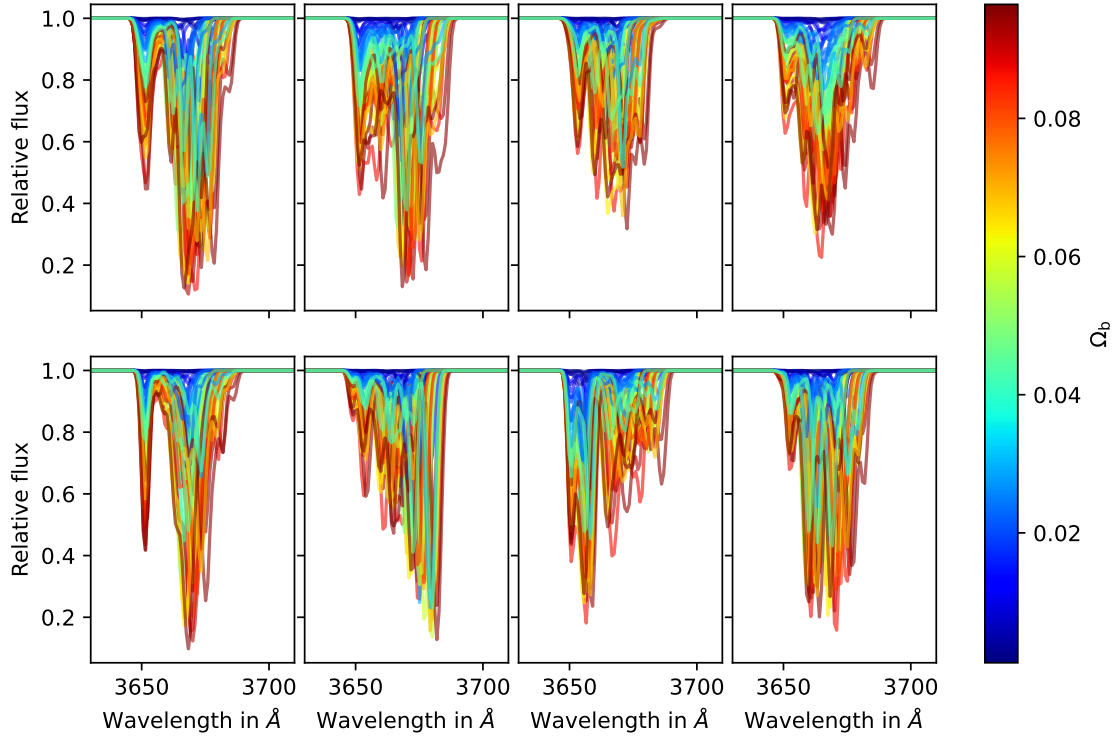
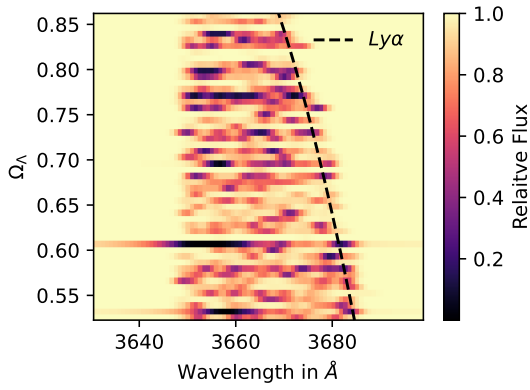
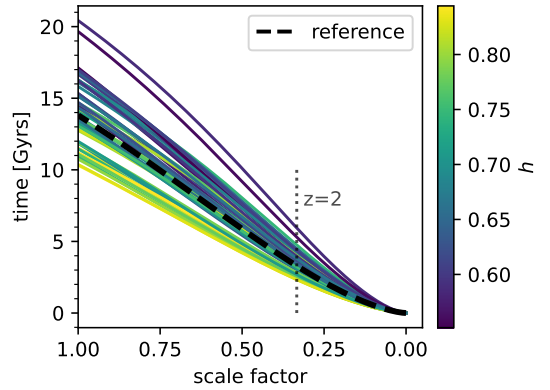


Figure 9: Example spectra, where each subplot shows spectra along the same line of sight for all 50 different boxes. A clear dependence of the depth and width of absorption features on Ω_b is visible.



(a) 2D Stack of spectra from the same lines of sight for all 50 boxes.



(b) Time vs redshift for all simulated boxes

Figure 10: 2D visualization of spectra along the same line of sight and visualization of the age of the universe. In (a) the clear trend for spectra from boxes with higher Ω_Λ to produce shorter spectra can be seen. In (b) the different ages of the universes vs scale factor can be seen. At $z = 2$ the boxes range from around 2 Gyrs to 6 Gyrs in age.

The dependence can actually be derived theoretically, by calculating how much redshift the light experiences traveling through the box. For the full derivation see Appendix 6.3. For the difference in redshift, we find

$$\Delta z = z + 1 - \frac{1}{\left(\frac{\Omega_m}{\Omega_\Lambda}\right)^{1/3} \sinh^{2/3} \left(\frac{3H_0 \sqrt{\Omega_\Lambda} d_c}{2c(1+z)} - \operatorname{arsinh} \left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m(1+z)^3}} \right) \right)}, \quad (8)$$

where d_c is the comoving box size, c the speed of light and z the initial redshift of the box. Knowing the difference in redshift, one can shift the Lyman- α wavelength by $(z + 1)$ and $(z + \Delta z + 1)$ respectively, and obtain the width w_s of the spectrum. Given this width, we can now simply shift spectra by $n \times w_s$ and add the optical depths $\tau_n(\lambda)$ together, giving us the full flux spectrum $F(\lambda) = \exp(\tau_1(\lambda) + \dots + \tau_n(\lambda))$. As the binning of wavelengths for the BOSS spectrograph is not constant, the flux values of the shifted spectra are associated with the closest available wavelength bin. This introduces a slight systematic error to the spectra, but is not of greater relevance, as binning does not deviate greatly over the chosen wavelength ranges. As we do not want to reduce the number of spectra in our training set, we produce 10,000 long spectra from the 10,000 short spectra per box by randomly shuffling short spectra together in the chosen wavelength ranges. A visualization of this can be seen in Fig. 11. Similar data augmentation strategies are common in machine learning applications. Another caveat is the fact that different wavelengths in the Lyman- α forest correspond to different redshifts. Filling longer regions of the spectra with simulations at redshift $z = 2$ will introduce yet another systematic error. We are thus limited by the length of spectra we can make under this approximation. We choose to augment the spectra from 3600 – 3950 Å, which corresponds to a redshift range of $\Delta z \approx 0.3$.

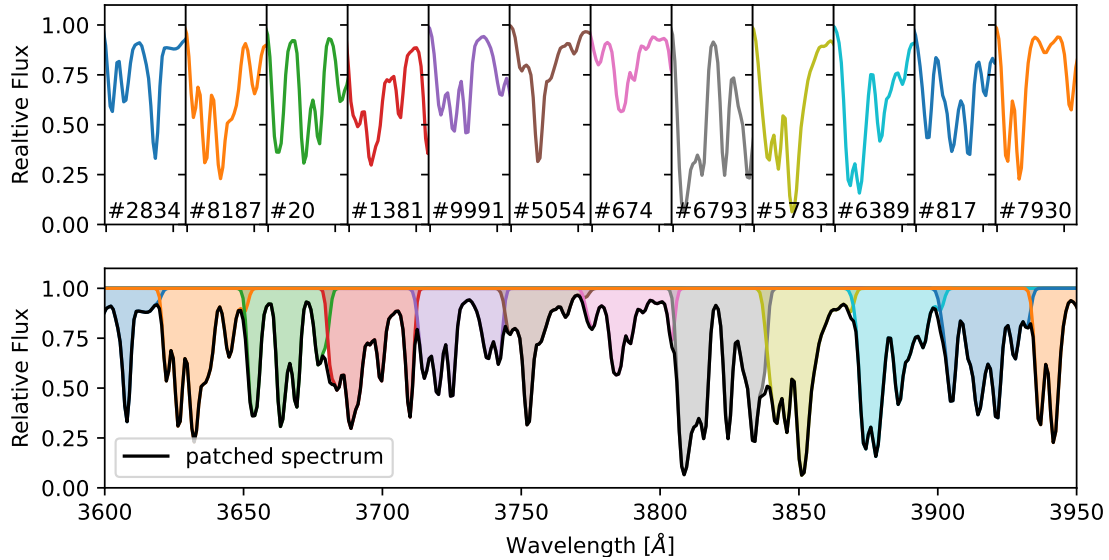


Figure 11: Visualization of the spectra augmentation. Different short spectra are randomly shuffled and patched together to make one long spectrum.

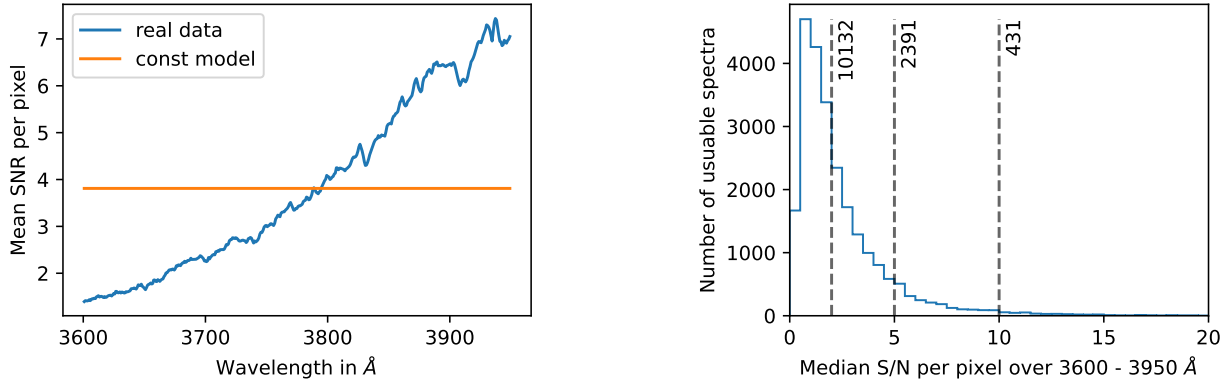
3 Machine Learning Methods

As motivated in Sec. 1.6, we choose the Transformer architecture for our inference network. First proposed by Vaswani et al. (2023), this architecture has revolutionized machine learning and was adopted in many different fields. It has lately been picked up by the Astrophysics community to process spectral data, which is available in large quantities thanks to large surveys such as SDSS (Lee et al. 2013) or DESI (Abdul Karim et al. 2025). As the Transformer architecture is relatively new, this is the first work adopting it for the purpose of cosmological parameter inference with Lyman- α forest spectra. We thus develop our own Transformer model, following the original design closely.

3.1 Preprocessing and Dataset Construction

Before implementing a model, one should think thoroughly about the data and how to preprocess it for training. There are two main objectives here: Firstly, we want to make the training data as realistic as possible, so that it reflects the inference inputs (SDSS spectra) as closely as possible. And secondly, we want to bring the data into a format, that is easy for the Transformer to process, while making sure that there is no information leakage between training and testing.

To increase the realism of our data, we implement two separate noise models. The first implementation based on the naive assumption of a constant Gaussian noise throughout the entire spectrum. This is our base model, as it is not specific to any instrument or dataset, and we thus train and test our pipeline with this model primarily. The second noise model is tailored to the SDSS DR 9 Lyman- α forest spectra catalog (Lee et al. 2013). For creating it, we first download the entire catalog and filter out spectra with NaN or infinite values in the range of 3600 – 3950 Å. Then we compute the median Signal-to-noise ratio (SNR) for every spectrum and index the spectra with this information. A histogram of the usable spectra’s median SNR be seen in Fig. 12b. Ideally, we would only use high SNR spectra, however, the number of spectra above any given SNR is limited and drastically decreases with higher SNR values. We thus choose to make three datasets for three different threshold values (2, 5, 10). In these datasets, we include all spectra with an SNR greater or equal than the threshold value.



(a) Mean Signal-to-noise per pixel for spectra in the SDSS catalog with a median SNR > 2 over the entire range.

(b) Median Signal-to-noise distribution of the usable spectra in the SDSS catalog for our work.

Figure 12: Statistics of Signal-to-noise (SNR) distributions for the spectra in the SDSS catalog. In (a) the mean SNR per pixel over all spectra is shown. In (b) the median SNR per spectrum over all pixels is shown.

However, the observed spectra do not simply follow the crude assumption of a constant SNR throughout

the entire wavelength range. They rather have dramatically lower SNRs for the lower wavelength range and higher SNRs for higher wavelength ranges (see Fig. 12a).

To improve the noise model from constant Gaussian throughout the entire spectrum to a more accurate one, we instead use the mean SNR value per wavelength bin throughout all selected spectra from the SDSS catalog (see Fig. 12a). A comparison between the noise models can be seen in Fig. 13.

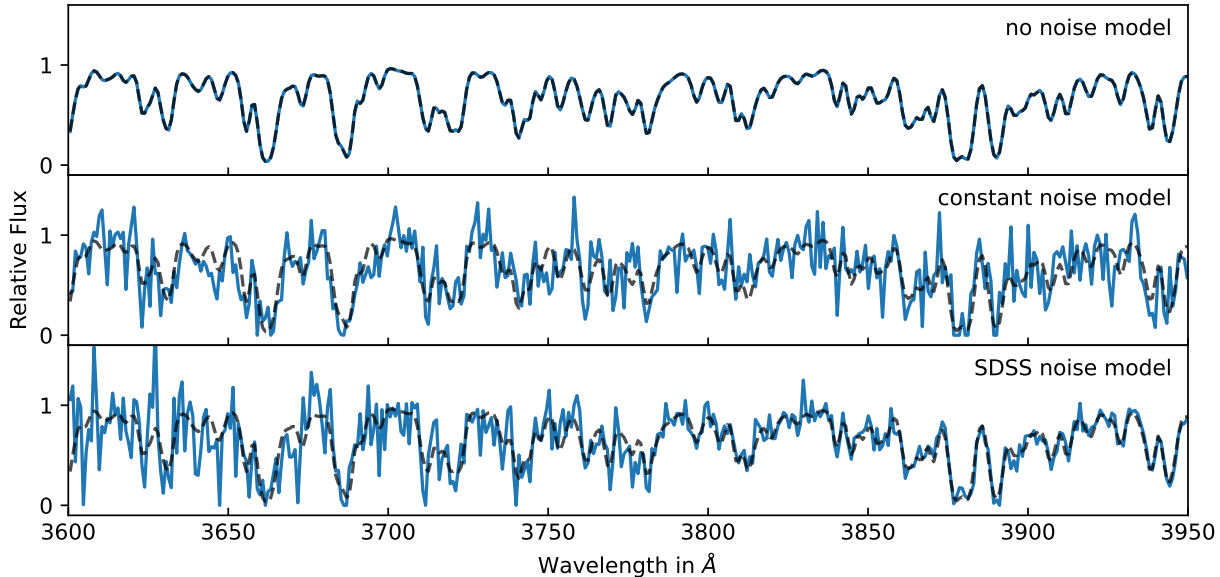


Figure 13: Comparison of the different noise models. In the upper plot no noise is added to the synthetic spectrum. In the middle plot a constant random Gaussian noise with Signal-to-noise (SNR) 5 is added. In the lower plot the mean SNR ratio per pixel of the SDSS catalog spectra with median SNR > 5 is assumed. In the lower plot, the SNR being below 5 for low wavelengths and above 5 for higher wavelengths can be clearly seen.

It is common practice to standardize the input data in machine learning applications. We, however, choose to not do so, as our spectra are already limited to values roughly in the range of $[0, 1]$. Additionally, the absolute values are relevant to the inference, because the depth of absorption features carries information about the density and physical state of hydrogen gas. We thus make datasets for the Transformer model by combining the noisy synthetic spectra X with the ground truth about the simulated cosmology $y = (\Omega_m, \Omega_b, \Omega_\Lambda, h)$. It is important to note, that we do standardize the y data, because the different ranges of cosmological parameters might induce a bias to the model.

We split the available (X, y) tuples into three distinct datasets with a Train/Validation/Test split of 70/15/15 percent. The model is trained on the majority of spectra (training set), while being evaluated on the unseen validation dataset for every trained epoch. This is done to test whether the network is just remembering the training data, or actually generalizing to unseen data. If the loss on the training dataset and the loss on the validation dataset drift apart, this is usually a sign of overfitting. Because the information of both the training and validation datasets are then used for hyperparameter optimization (which we discuss in Section 3.4), we need a third and separate dataset to then test the final performance of the model (the test set).

As we already have a low number of sample points of the parameter space (50 samples), we include all 50 boxes in training. To archive separate datasets from the 50 boxes, we split the boxes into three different regions with the same split as for the datasets of 70/15/15 percent. For each dataset we only sample lines of sight from one region, and thus reduce the chance of sampling the same halo in different datasets. As this

method might still have a low amount of information leakage between the training and test sets due to the regions being adjacent, we run a 51st reference box with cosmological parameters aligning with the Planck 2015 measurements (Planck Collaboration et al. 2016). This box has never been seen during any steps of training and therefore can be used as a final test of the quality of inference.

In total, we compile six different datasets with differing SNR values and noise models and train one model for each dataset with the same hyperparameters. An overview over the models and datasets can be seen in Table 2.

Table 2: Overview of dataset configurations and models trained

Model name	SNR	Noise model	simulation suite	initial conditions
C2	2	constant noise	CLIMB high	fixed
C10	10	constant noise	CLIMB high	fixed
C100	100	constant noise	CLIMB high	fixed
R2	2	SDSS realistic noise	CLIMB high	fixed
R5	5	SDSS realistic noise	CLIMB high	fixed
R10	10	SDSS realistic noise	CLIMB high	fixed
R10-128	10	SDSS realistic noise	CLIMB low	fixed
R10-128-var	10	SDSS realistic noise	CLIMB low	varied

3.2 Transformer Architecture

As illustrated in Fig. 14, the first component of our Transformer consists of a linear embedding layer combined with positional encoding (PE). In our setup, each flux value corresponding to a wavelength of the input spectra is treated as an individual token and is embedded independently. The linear embedding projects each token into a d_{model} -dimensional space, which we refer to as the feature space. In this space, the network tries to learn a meaningful representation of the spectral data. Each spectrum with length L is thus projected into a space with dimension (L, d_{model}) . Treating every flux value as a token, however, ignores the positional information of the corresponding wavelengths. To inject back the information about the relative (wavelength) position of tokens to each other, we add a sinusoidal positional embedding:

$$PE_{(\text{pos}, 2i)} = \sin(\text{pos}/10000^{2i/d_{\text{model}}}) \quad (9)$$

$$PE_{(\text{pos}, 2i+1)} = \cos(\text{pos}/10000^{2i/d_{\text{model}}}), \quad (10)$$

where pos is the position in the spectrum and i is position in feature space. The idea behind this PE is to encode each feature dimension with its unique frequency. The position in the spectrum is then just a phase in this Fourier basis of feature space. This embedding has the advantage that for every fixed offset k , $PE_{\text{pos}+k}$ can be represented as a linear transformation of PE_{pos} (Vaswani et al. 2023), which is intrinsically easy for the network to learn. We also implemented Relative Positional Encoding (RPE), which lead to similar or slightly worse results. We thus use the standard PE in all our models.

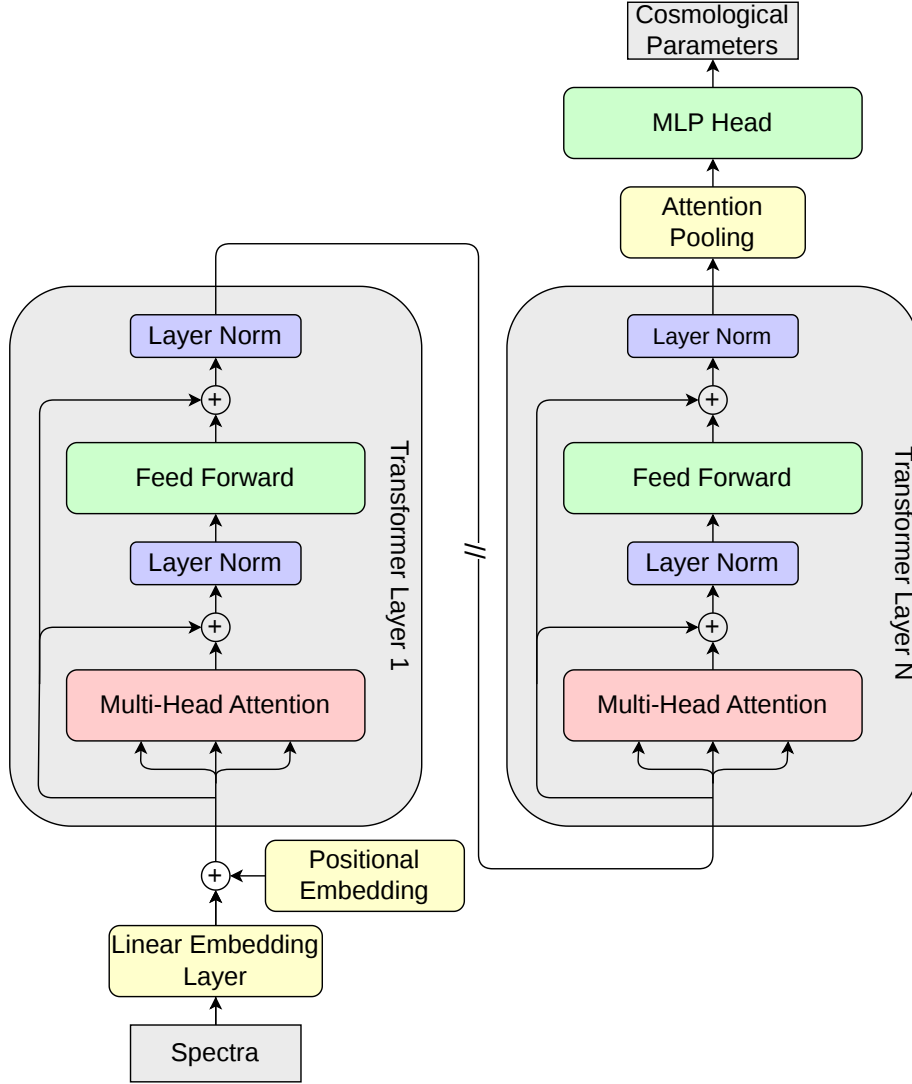


Figure 14: Flow chart of Transformer architecture used in this work.

The next and arguably most important component of the Transformer are the Transformer layers. They consist of a Multi-Head Attention and a Feed Forward block with Residual Connections and Layer Normalizations. For Multi-Head Attention splits the feature space is split into subspaces, which are then processed independently by different Attention Heads. This allows the network to attend to different types of relationships between the input tokens simultaneously. The output of all Attention Heads is then concatenated and given to the fully connected feed forward network, which consists of one input, one hidden and one output layer. It projects each token from the d_{model} -dimensions input space into a d_{ff} -dimensional space (which is usually larger than d_{model}) and then back to d_{model} dimensions individually. This Feed Forward network learns a representation of the data in the feature space. This learning can be seen as updating the representation of the tokens in feature space with the output from the Attention Heads. After both blocks (Transformer layers and the feed forward network) a Residual connection is added, and a layer normalization is applied. This is done to stabilize training and facilitate the flow of gradients through the model. The overall depth of the model, i.e. the number of Transformer layers num_{layers} , as well as the number of attention

heads num_{heads} and the dimensionality of the hidden representations d_{ff} , are all architectural hyperparameters. These control the expressive capacity of the network and are determined through hyperparameter optimization described in Section 3.4.

The final component of our Transformer constitutes the Multi-Layered Perceptron (MLP) Head. This MLP head is what makes the actual prediction and maps its input into the parameter space. However, for stabilization and convergence purposes, we do not feed the $L \times d_{model}$ dimensional output of the last Transformer layer into this MLP directly. Rather, we apply one Attention pooling layer first. This distills the important information by learning a set of weights that determine the relative importance of each token for the final prediction. Mathematically, it learns a query vector that attends to the sequence of encoded token representations and computes a weighted sum, allowing the model to emphasize the most informative spectral regions. The final MLP itself is then just one fully connected layer reducing d_{model} -dimensions to $dim(\Theta)$ -dimensions, where Θ is the parameter space.

It is worth noting that the Transformer architecture naturally supports input masking, which can be used to exclude wavelength bins with bad pixels in the spectra. In principle, this would allow the inclusion of observational spectra with incomplete or corrupted regions. However, incorporating such masking consistently would also require modeling these effects in the training simulations to avoid introducing a mismatch between training and inference data. As this lies beyond the scope of this thesis, we restrict ourselves to fully valid spectra and do not implement input masking. Nevertheless, extending the framework to account for masked inputs is a possible direction for future work.

3.3 Training

The Transformer model is trained to learn a mapping between the spectral data and cosmological parameters by minimizing a supervised regression loss function. For the loss function we choose the mean squared error (MSE), which penalizes large deviations between the prediction and truth especially strongly. This loss function provides a stable and well-understood optimization target for continuous parameter estimation. The model weights are updated using the AdamW optimizer, which combines adaptive gradient-based updates with decoupled weight decay regularization. The weight decay regularization biases the network towards smoother mappings which often leads to a better generalization capability, particularly in large neural networks.

In addition to weight decay, several further measures are taken to mitigate overfitting. This is particularly important in our setting, as we only have a relatively low number of sample points for the cosmological parameter space. To improve generalization, dropout layers are applied at multiple stages of the architecture, including the embedding layer, the multi-head attention blocks, and the feed-forward networks within the Transformer layers.

Training is performed using mini-batches of the dataset, allowing gradients to be computed on subsets of the full training set at each optimization step. This approach improves computational efficiency, as modern GPUs are highly optimized for parallel matrix operations, while also stabilizing training by reducing noise in the gradient estimates. The model is trained for a total of six epochs, as the training and validation loss curves reach a plateau around this point and further training does not lead to noticeable improvements while increasing the risk of overfitting. An example loss curve is shown in Fig. 15.

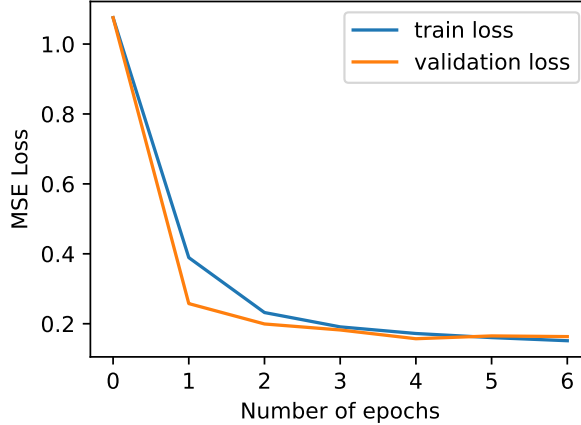


Figure 15: Training and validation loss during optimization for the C100 model. Both curves have reached a plateau after six training epochs.

The exact values of the training and architectural hyperparameters are determined through a dedicated hyperparameter optimization procedure, which is described in the following Section (3.4). In its final configuration, training the model on the full dataset of 500,000 spectra requires approximately one hour on a single Nvidia A100 GPU.

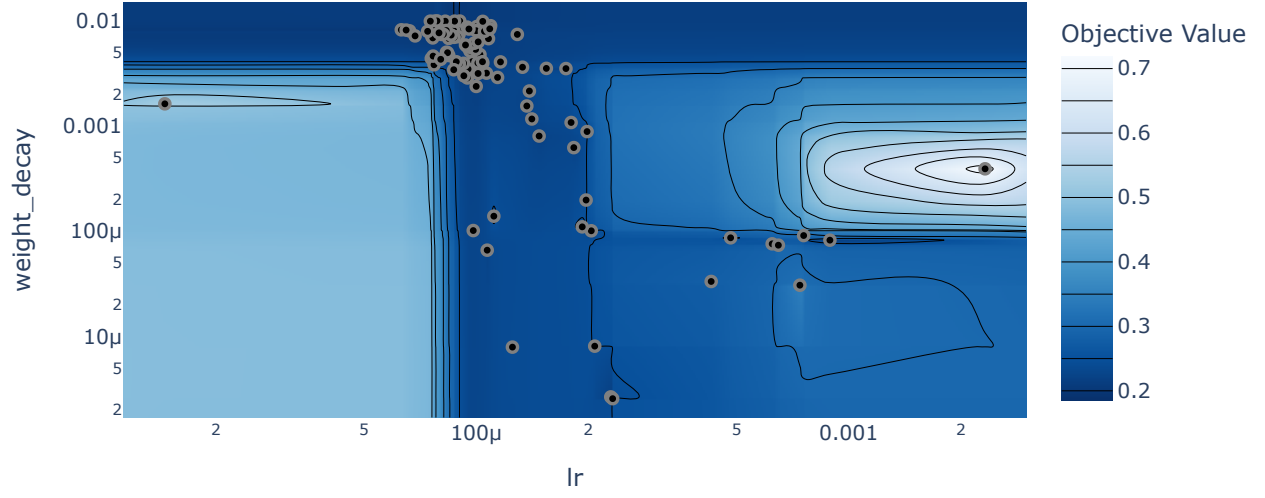
3.4 Hyperparameter Fine-Tuning

We perform hyperparameter fine-tuning (or hyperparameter optimization) with the Optuna package, which does not simply brute force all possible combinations, but rather employs efficient sampling strategies to iteratively focus on the most promising hyperparameter space regions. Under the hood Optuna uses the Tree-structured Parzen Estimator (TPE) algorithm, which creates probabilistic models for both well performing and poorly performing regions of the hyperparameter space based on previous trials. The algorithm then preferentially samples regions of the hyperparameter space that are expected to yield improved performance.

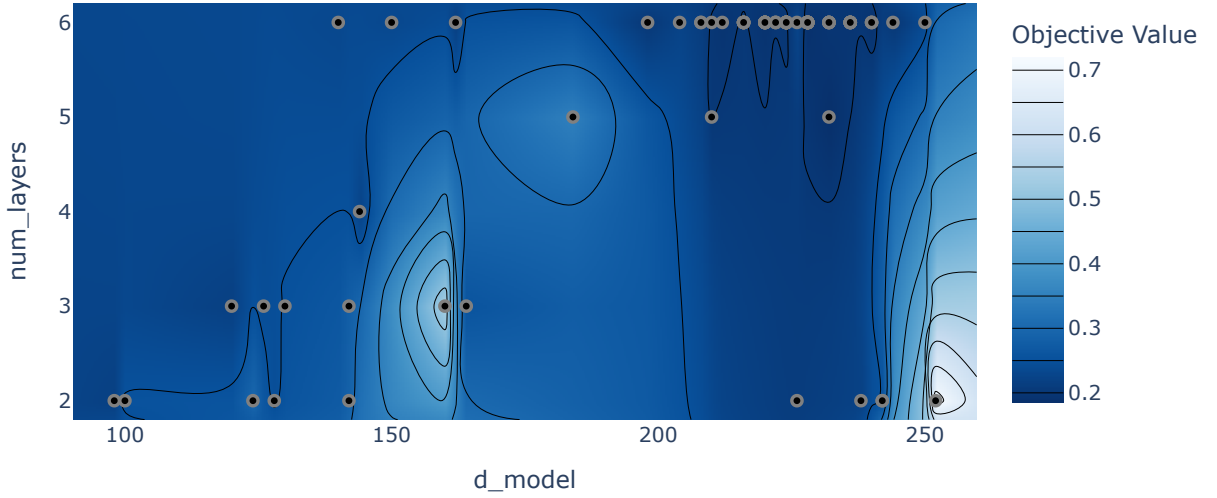
To further increase efficiency, Optuna has a pruning mechanism, which stops poorly performing trials early in the training process. During the training of each trial, it monitors intermediate validation loss values and stops the trials that are unlikely to outperform the current best configuration early. This strongly reduces unnecessary computation and allows the available resources to be used for more promising hyperparameter configurations. Additionally, Optuna supports parallel execution of trials, which makes it possible to train multiple models simultaneously.

In this work, we optimize eight different hyperparameters that include both architectural parameters of the Transformer and training-related parameters. The objective function used during optimization is the MSE of the forward passed validation dataset. In total, we approximately use 2000 trials in the optimization process to sample this eight-dimensional space. Examples of two-dimensional slices of the hyperparameter space and its sampling as well as the interpolated loss landscape can be seen in Fig. 16.

The final product of this hyperparameter optimization procedure and our final model configuration as well as the ranges and explored hyperparameters can be seen in Table 3. In its final configuration, our model has almost 4 million trainable parameters.



(a) Learning rate vs weight decay slice of the hyperparameter space



(b) Feature space dimension d_{model} vs number of Transformer layers num_{layers} slice of the hyperparameter space

Figure 16: Two example contour plots of the hyperparameter optimization. In (a) the loss landscape of learning rate and weight decay can be seen. Most feasible trials cluster in one minimum. In (b) the loss landscape of d_{model} vs num_{layers} can be seen. This landscape seems to have multiple regions with local minima and thus the feasible trials are spread out more over the landscape.

Table 3: Parameter ranges for hyperparameter optimization and results

Hyperparameter	Search Range	Final Value
d_{model}	[64, 256]	232
num_{heads}	{2, 4, 6, 8}	4
num_{layers}	[2, 6]	5
d_{ff}	[64, 2048]	1160
learning rate η	[1e-5, 5e-3]	8.308e-05
weight decay λ	[1e-6, 1e-2]	0.0080898
dropout	[0.0, 0.1]	0.002254
batch size	{16, 32, 64, 128, 256}	16

4 Results

4.1 Inference Performance on Dataset with Constant Noise Model

We first test the performance of our method and pipeline on the constant-noise models and find that they are clearly capable of retrieving information about the four cosmological parameters from the Lyman- α forest spectra. This is done by forward passing the respective test datasets through the three trained models C2, C10, and C100, obtaining $p(\hat{\theta} \mid \theta_*)$ for each parameter. This allows us to evaluate both the bias and variance of the predictions for each true parameter value θ_* . The inference results for the C10 model are shown in Fig. 17. The predictions follow the true parameter values closely for most of the explored parameter space. A complementary visualization of the distributions $p(\hat{\theta} \mid \theta_*)$ is shown in Fig. 18, where the peaks of the distributions appear as bright spots centered on the true parameter values.

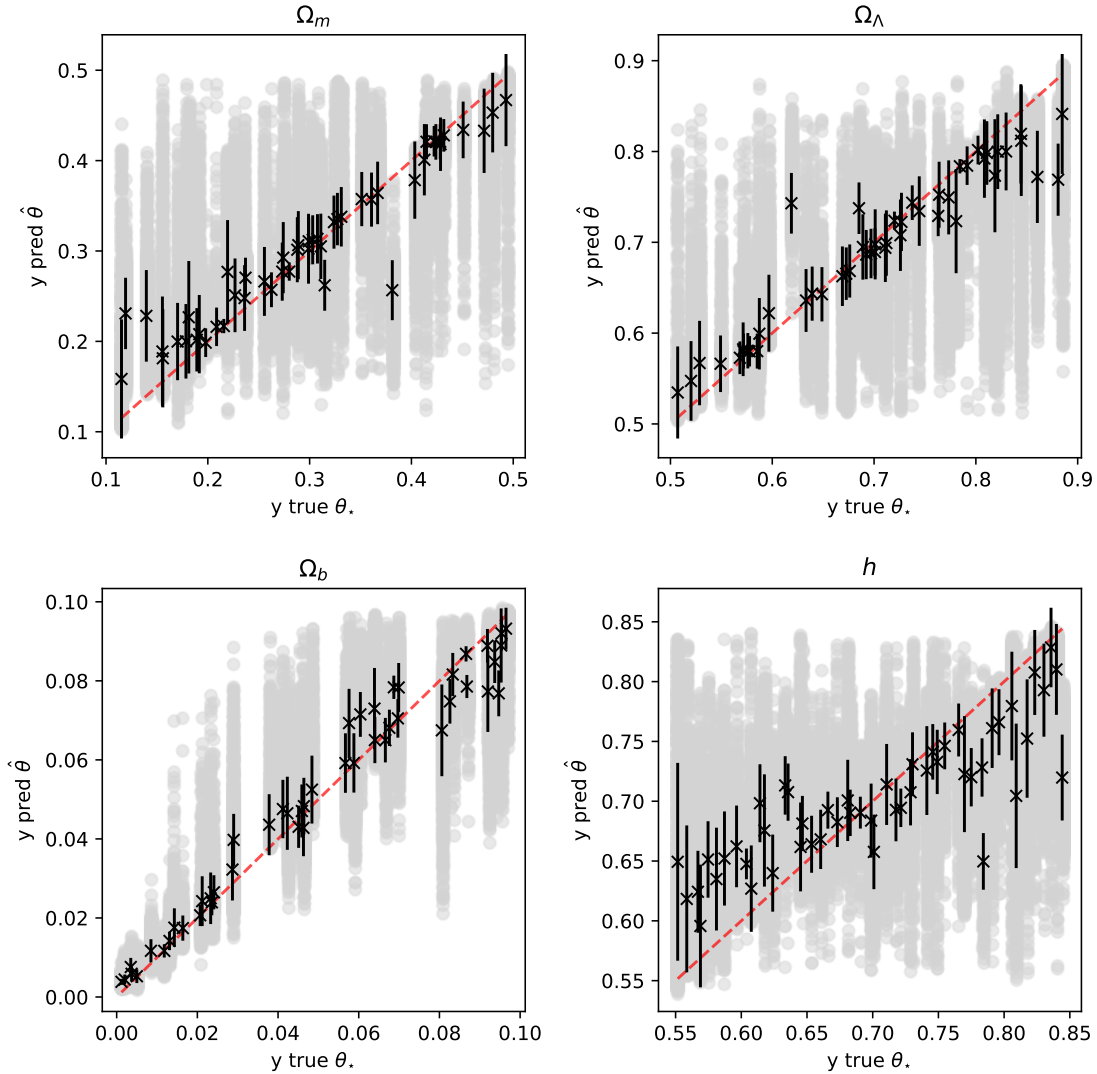


Figure 17: Inference results of the C10 model on its test dataset. The predictions follow the true values closely for Ω_m , Ω_Λ and Ω_b . For h there is a visible bias to overpredict the lower end of the parameter range and to underpredict the upper end of the parameter range.

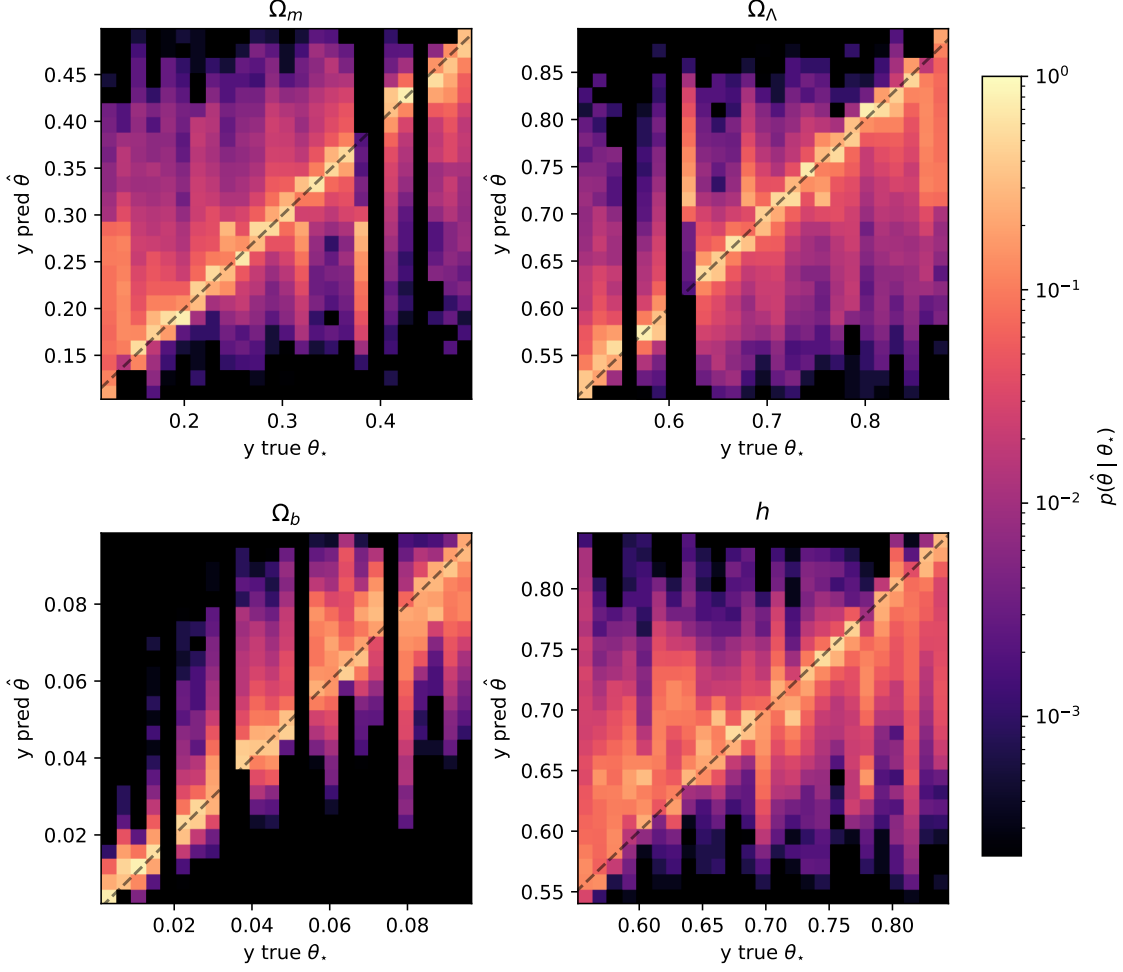


Figure 18: Complementary figure for the inference results of the C10 model on its test dataset. In this figure, the predicted distributions can be seen as color.

Further insight can be obtained from slices through the parameter space (Fig. 19), which show that the predicted parameter distributions generally peak near the correct ground-truth values. The parameter that appears most difficult to infer is h , which shows a systematic bias toward overestimation at the lower end of parameter space and underestimation at the upper end. This behavior is not unexpected, since constraints on h are typically derived from simulations or measurements over much larger distances than 25 Mpc, either by simulating a larger cosmological volume or by including multiple snapshots at different redshifts in the spectra creation.

While the predictions are accurate for most true parameter values θ_* , a small number of outliers appear in the inference results. These outliers likely correspond to cases, where one of the other parameters takes on extreme values. For example, boxes with extremely low baryon densities ($\Omega_b \approx 0$) contain very little absorbing material, resulting in spectra with almost no absorption features. In these situations, the spectra carry very limited information about the underlying cosmological parameters, making accurate inference almost impossible.

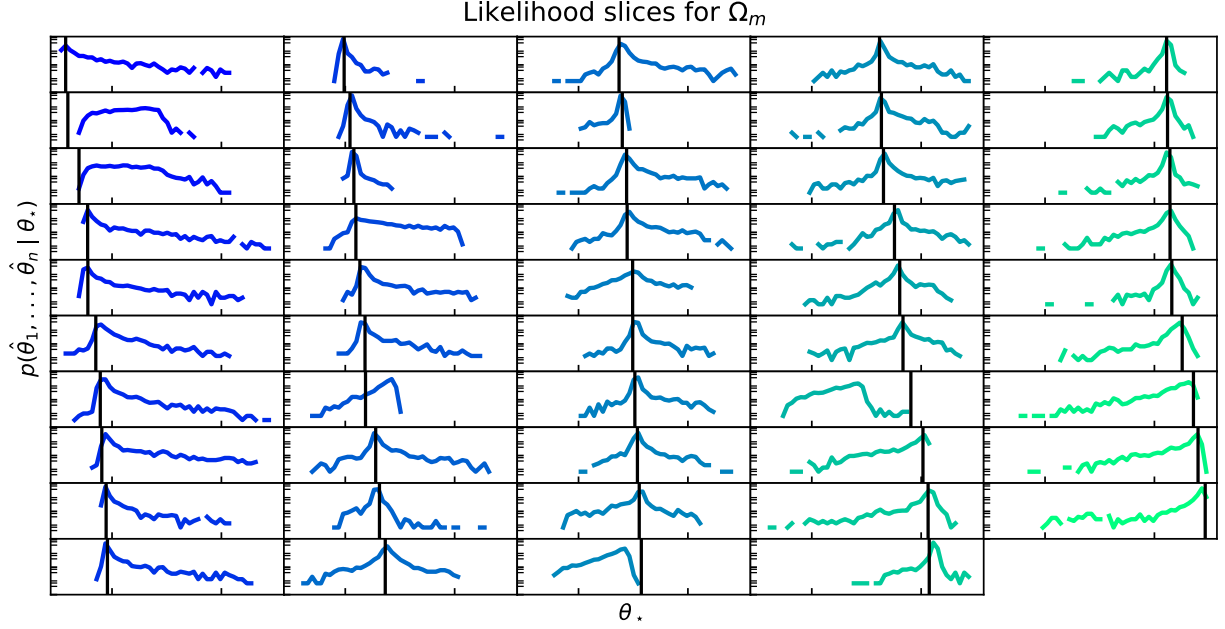


Figure 19: Log Likelihood slices of the predictions made by the C10 model on the test dataset for Ω_m . Most distributions have a peak around the correct values.

The model also preserves the flat universe condition present in the training data. We test this by computing the curvature parameter $k = \Omega_m + \Omega_\Lambda$ for each inference point. As shown in Fig. 20, the predicted values consistently find $k = 0$. We find this behavior for all eight models. This behavior is expected, since all training samples enforce a flat universe. Nevertheless, the network is not explicitly constrained to satisfy this relation and could, in principle, predict combinations of Ω_m and Ω_Λ corresponding to non-zero curvature.

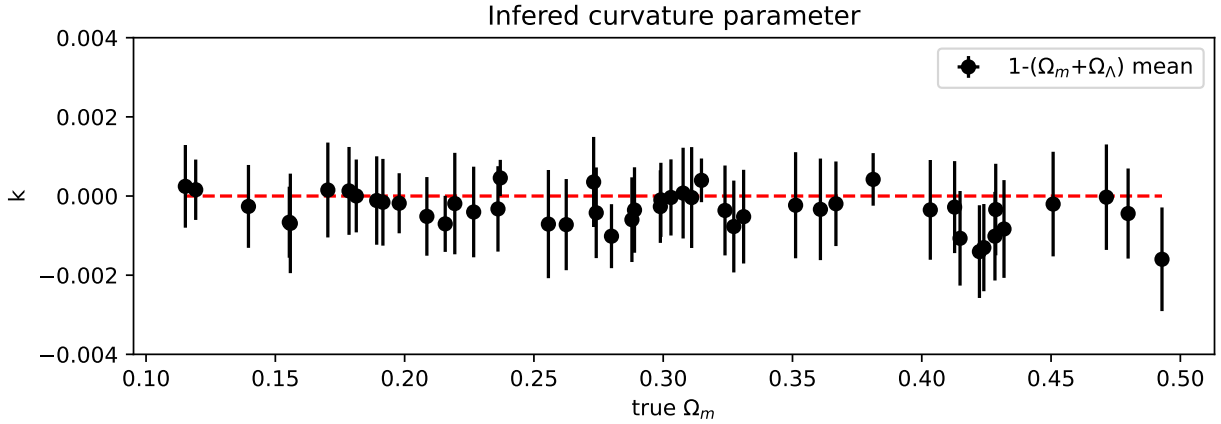


Figure 20: Indirectly inferred curvature parameter of the C10 model on its test dataset. The model predicts $k = 0$ consistently throughout the entire parameter space.

Inference on the reference boxes with Planck cosmology further demonstrates the accuracy of the model predictions. The resulting parameter distributions are shown in Fig. 21. The predicted distributions for Ω_m and Ω_Λ show sharp peaks centered on the true values, while the distributions for Ω_b and h are noticeably wider. The broader distributions for these parameters reflect the fact that they are more difficult to constrain

within the current simulation setup. In particular, h is expected to be poorly constrained given the relatively small simulation volume. The wider distribution for Ω_b likely arises because this parameter has only a weak effect on the expansion history of the universe, meaning that the network must rely primarily on properties of the absorption features themselves, such as their number and depth. These properties can vary significantly even within a single simulation box, leading to a larger spread in the inferred values.

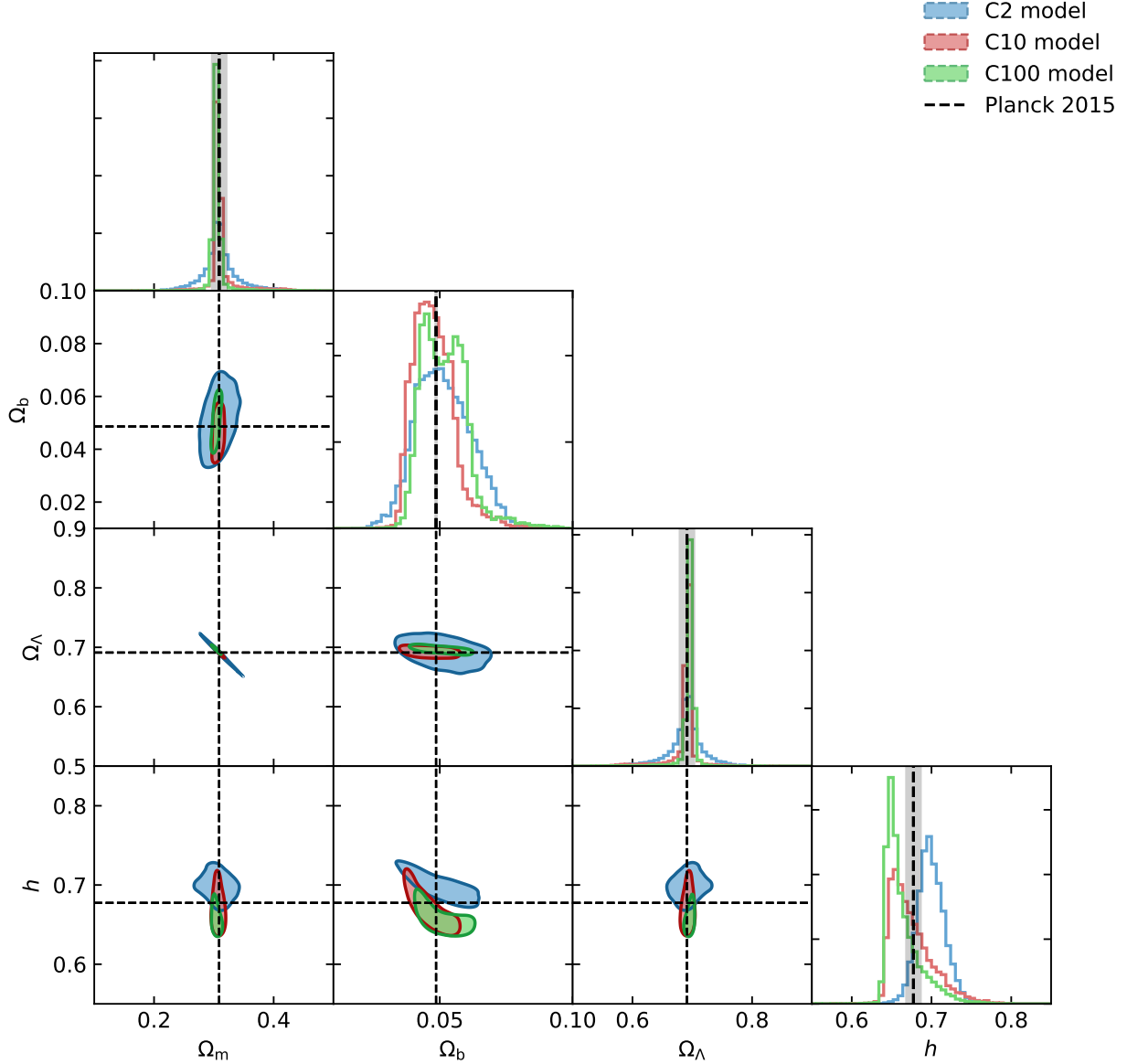


Figure 21: Inference results of the C2, C10 and C100 models on the Planck cosmology reference box. Ω_m and Ω_Λ are predicted accurately with sharp peaks, while Ω_b and h have wider distributions centered generally around the correct values.

The signal-to-noise ratio of the spectra significantly affects the accuracy of the parameter inference. To quantify this, we compute the bias and variance of the predictions by comparing the mean predicted values on the test set with the true parameters and measuring the corresponding 1σ intervals (Fig. 22). We find that both the variance and the bias increase as the noise level increases. This effect is especially noticeable for spectra with $\text{SNR} = 2$, where the bias can make up a substantial fraction of the predicted value in the

lower regions of parameter space. This demonstrates that noise directly limits the achievable precision of cosmological parameter inference on the Lyman- α forest spectra.

Overall, the models are able to recover the cosmological parameters with good accuracy across the explored parameter space. The primary limitations arise for h , which shows systematic biases likely related to the limited simulation volume, and for low signal-to-noise spectra where noise significantly increases both variance and bias.

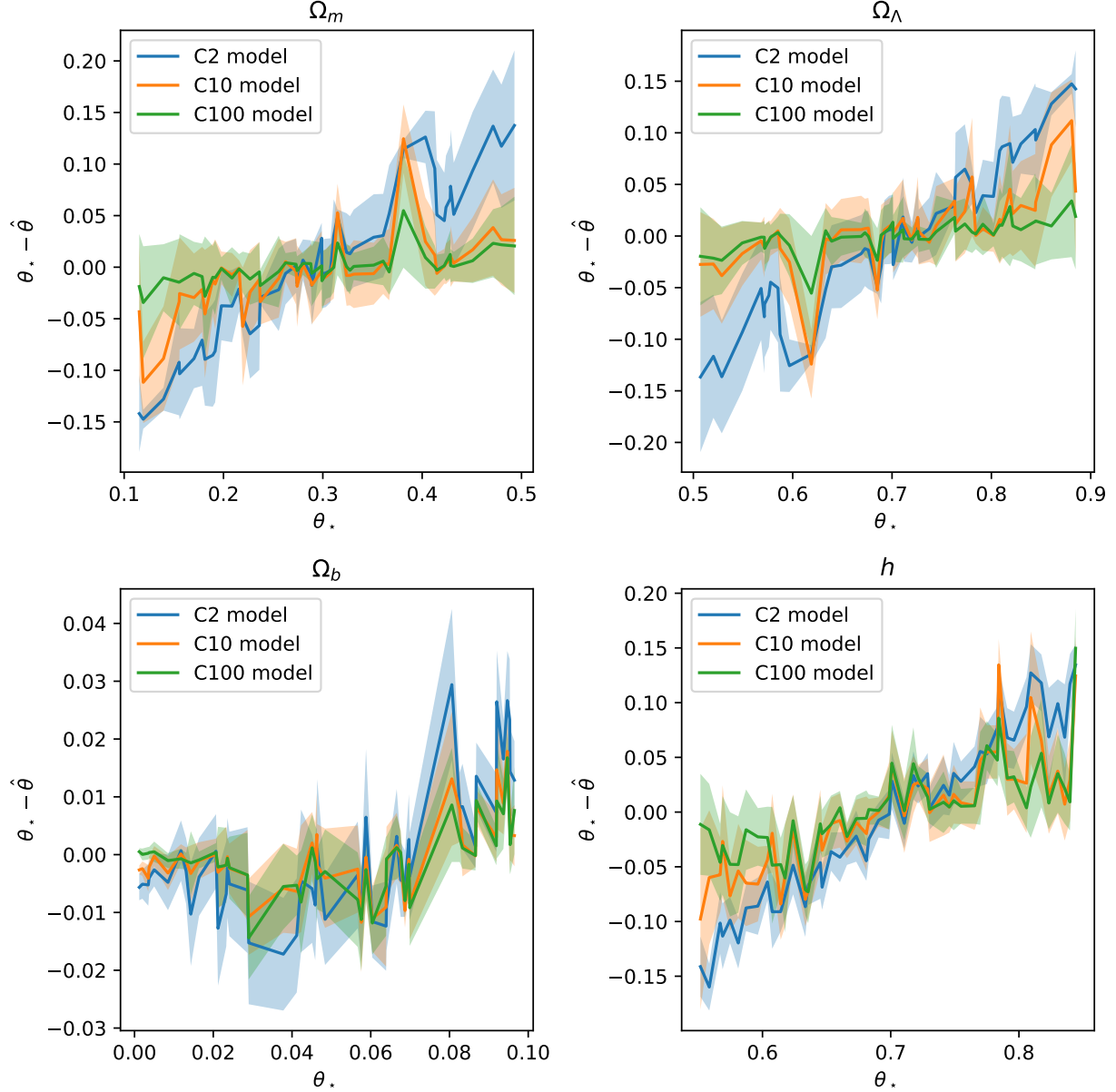


Figure 22: Bias of the constant noise models over the parameter space. It is clearly visible that the lower signal-to-noise models have both a higher bias and a greater variance.

4.2 Inference Performance on Dataset with Realistic SDSS Noise Model

The Transformer models trained on datasets with a realistic SDSS noise model (R2, R5, and R10) achieve a performance comparable to those trained on datasets with constant noise. This can be seen in Fig. 23, where the predictions of the R10 model closely follow the true parameter values for Ω_m , Ω_Λ , and Ω_b across almost all test samples, with only a few outliers. Similar to the constant noise models, the inference of h shows a systematic bias, where lower values tend to be overestimated while higher values are underestimated.

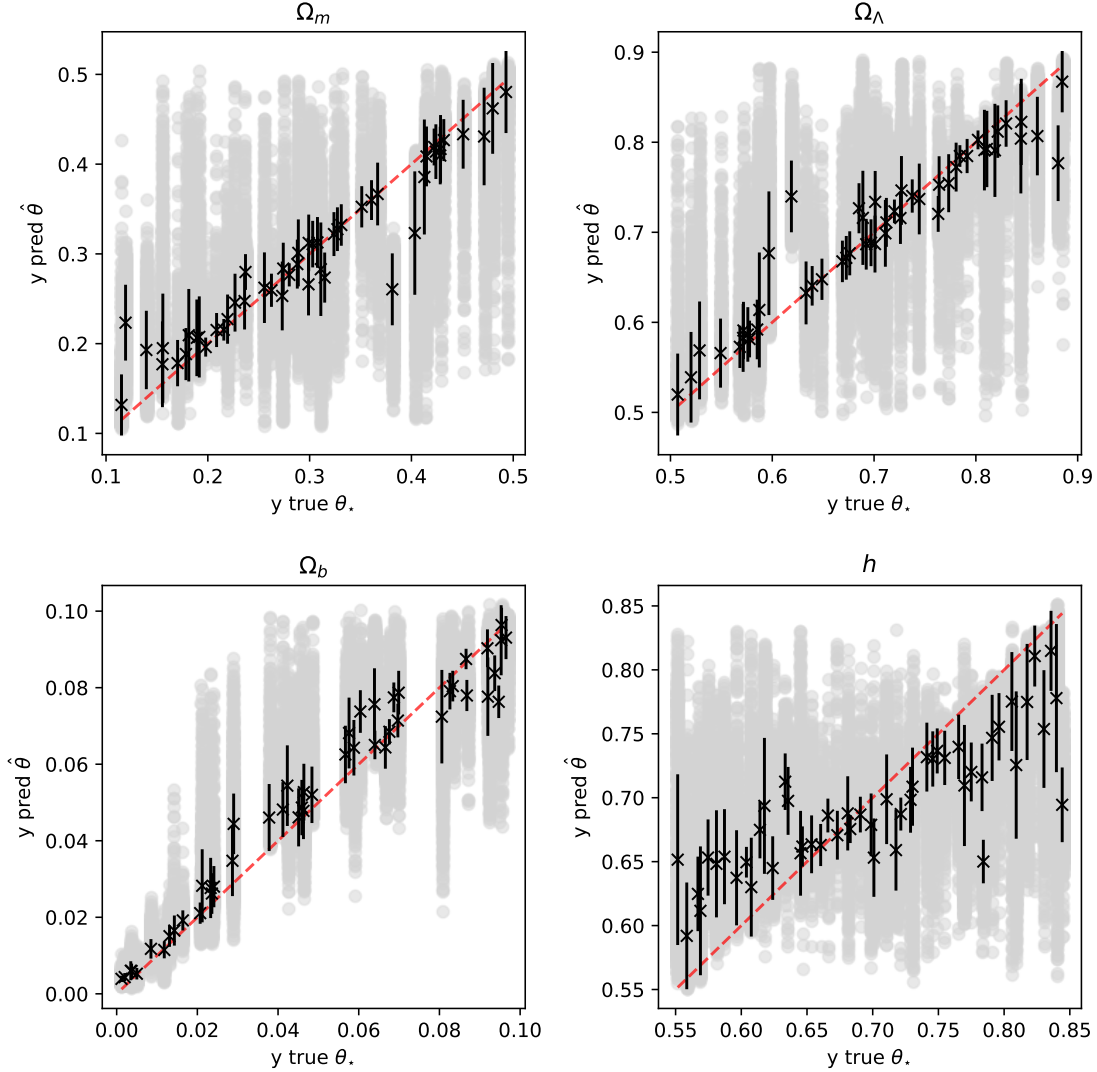


Figure 23: Inference results of the R10 model on its test dataset. Similar to the C10 model, the predictions follow the true values closely for Ω_m , Ω_Λ and Ω_b . For h there is a visible bias to overpredict the lower end of the parameter range and to underpredict the upper end of the parameter range.

Further evidence for the robustness of the model is obtained from inference on the reference box. The resulting parameter distributions trace the ground truth values accurately, as depicted in Fig. 24. In particular, the predicted distributions for Ω_m and Ω_Λ show sharp peaks centered on the correct values, with widths that are competitive with the uncertainties reported by the Planck measurements. In contrast, the distributions for

Ω_b and h are noticeably broader, although they remain centered around the correct values.

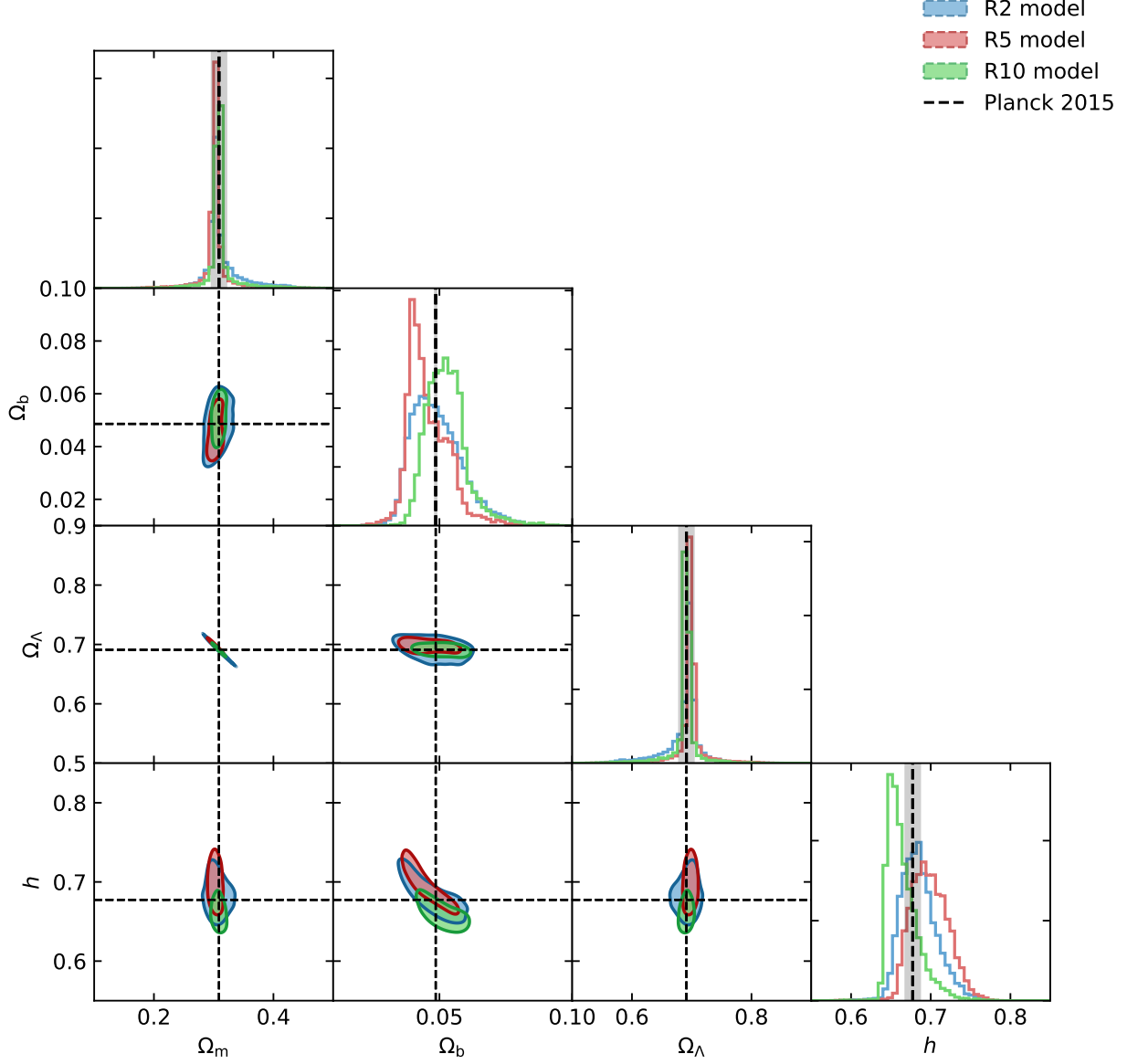


Figure 24: Inference results of the R2, R5 and R10 models on the Planck cosmology reference box. Similarly to the constant noise models, Ω_m and Ω_Λ are predicted accurately with sharp peaks here, while Ω_b and h have wider distributions centered generally around the correct values.

Interestingly, the dependence of the model bias on the SNR of the data is significantly smaller for the realistic SDSS noise model, as seen in Fig. 25. The tendency for the lower SNR models to have a higher bias is still visible, but much weaker compared to the constant noise models. This is likely due to the fact that the noise within the individual spectra vary over a wider range than the difference between the SNR thresholds of the three datasets. This also hints at the fact that the model uses information from the entire spectra in the inference process and does not simply focus on the high SNR regions.

These results indicate that, while the noise model can slightly influence the bias behavior, the choice of noise model only has a minor impact on the overall performance of the pipeline, provided that the noise properties

of the training and test datasets are consistent.

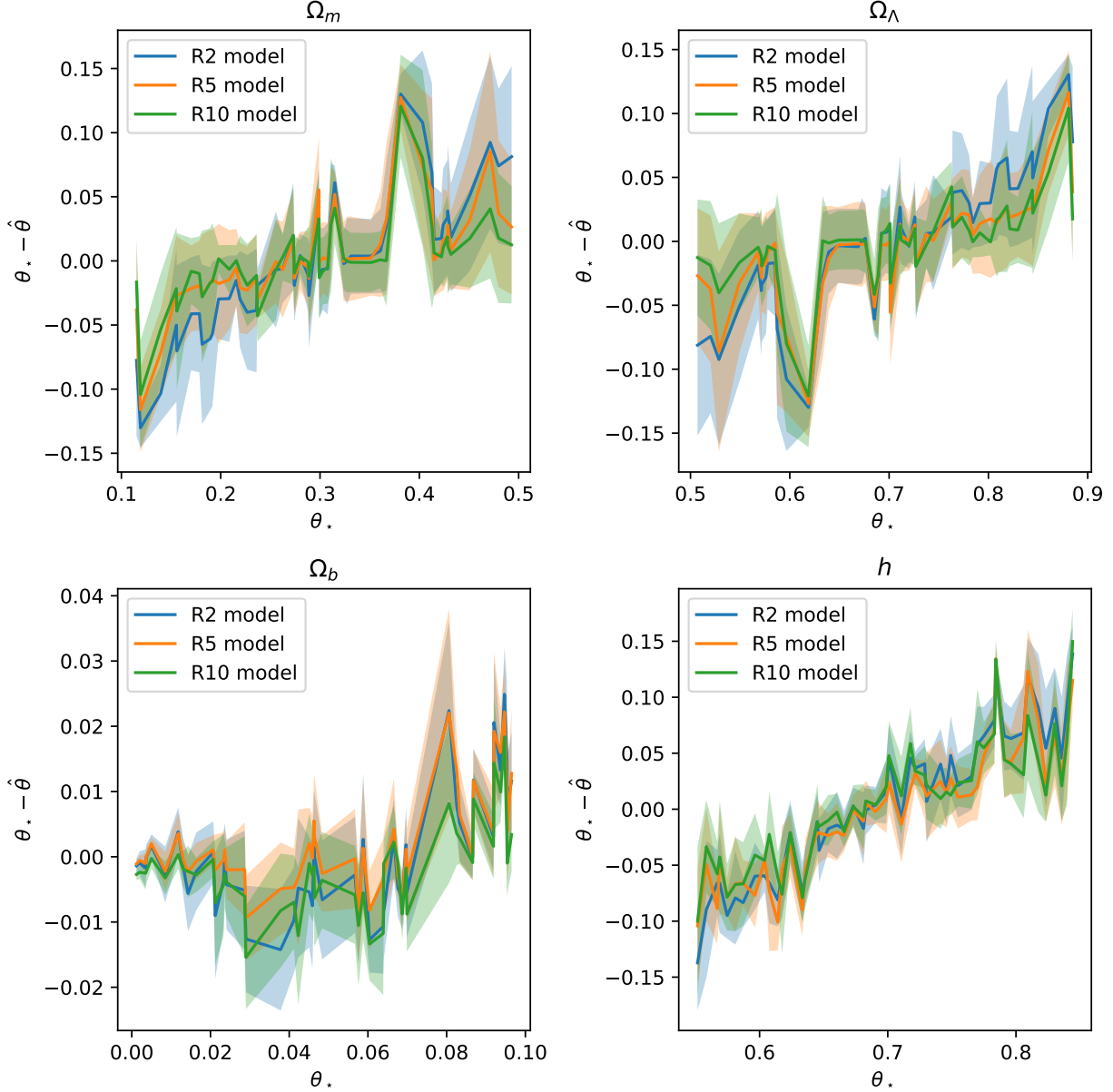


Figure 25: Bias of the realistic SDSS noise models over the parameter space. Contrary to the constant noise models, the SNR value does not seem to have a strong effect on the model bias and variance here.

4.3 Inference Differences Between Fixed ICs and Varied ICs with the CLIMB low Suite

Our choice of fixed initial conditions within the CLIMB high suite is unlikely to significantly influence the model performance. We test this by training the R10-128 model on the CLIMB low suite with fixed ICs and training the R10-128-var model on the CLIMB low suite with varied ICs. The inference results of both models on the same reference box can be seen in Fig. 26. The predicted distributions center around similar

values and generally have a similar shape. The inference performance for both models is noticeably lower than the rest of our models, however, this is most likely caused by the lower resolution in the simulations. We thus conclude that it is a valid choice to keep the initial conditions fixed between the different simulated boxes, and that this choice does not introduce a relevant bias to the pipeline.

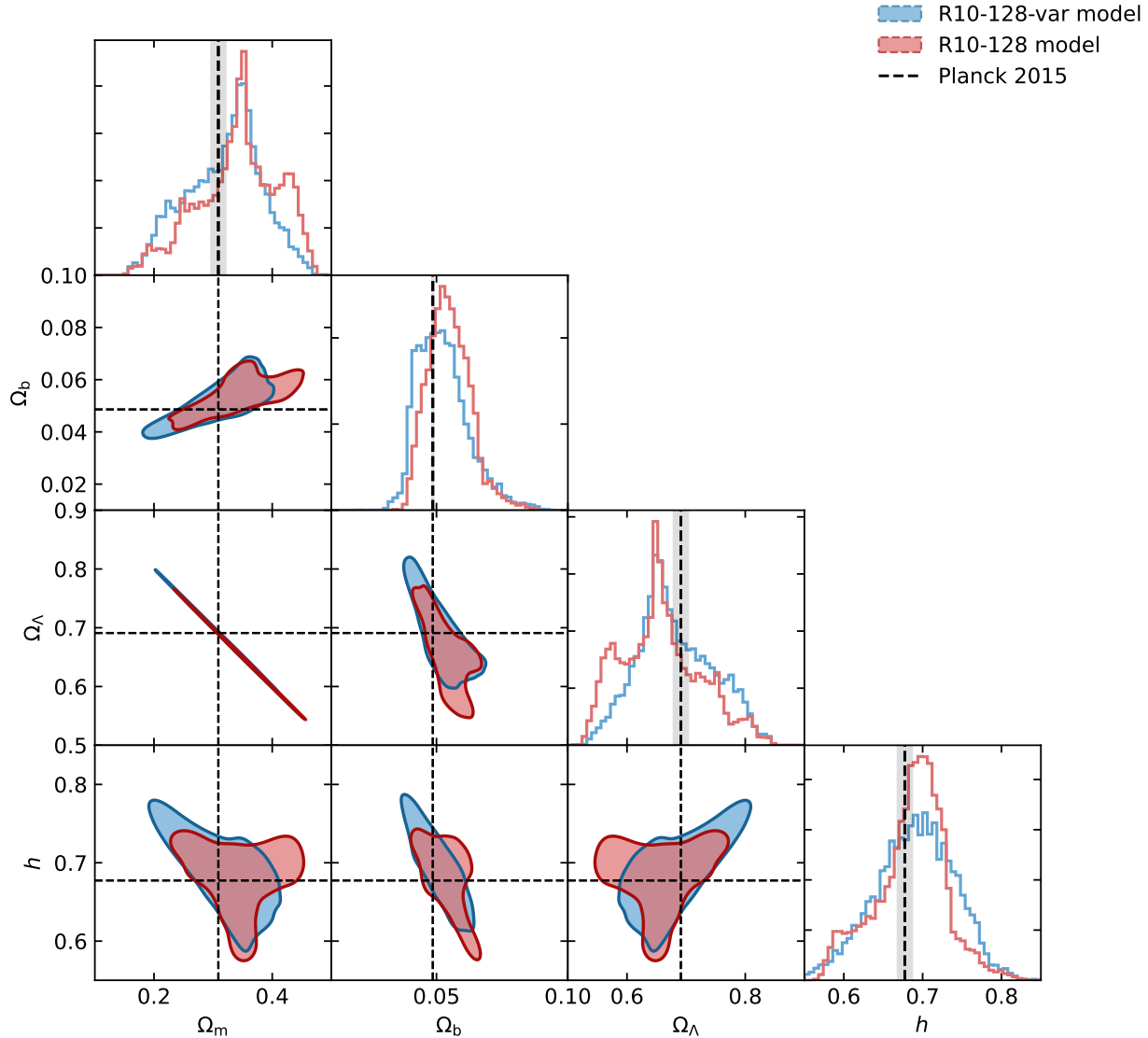


Figure 26: Inference results of the R10-128 and R10-128-var models on the same Planck cosmology reference box. The performance of the two models is comparable and does not deviate greatly for any parameter.

4.4 Inference on SDSS Spectra

Applying the pipeline developed in this work to observed Lyman- α forest spectra from the SDSS catalog, we find $\Omega_m = (0.29 \pm 0.04)$, $\Omega_b = (0.025 \pm 0.007)$, $\Omega_\Lambda = (0.72 \pm 0.04)$ and $h = (0.721 \pm 0.026)$. The inference is performed using the R2, R5, and R10 models trained on the corresponding training datasets, and combining their predictions. The resulting parameter distributions are shown in Fig. 27.

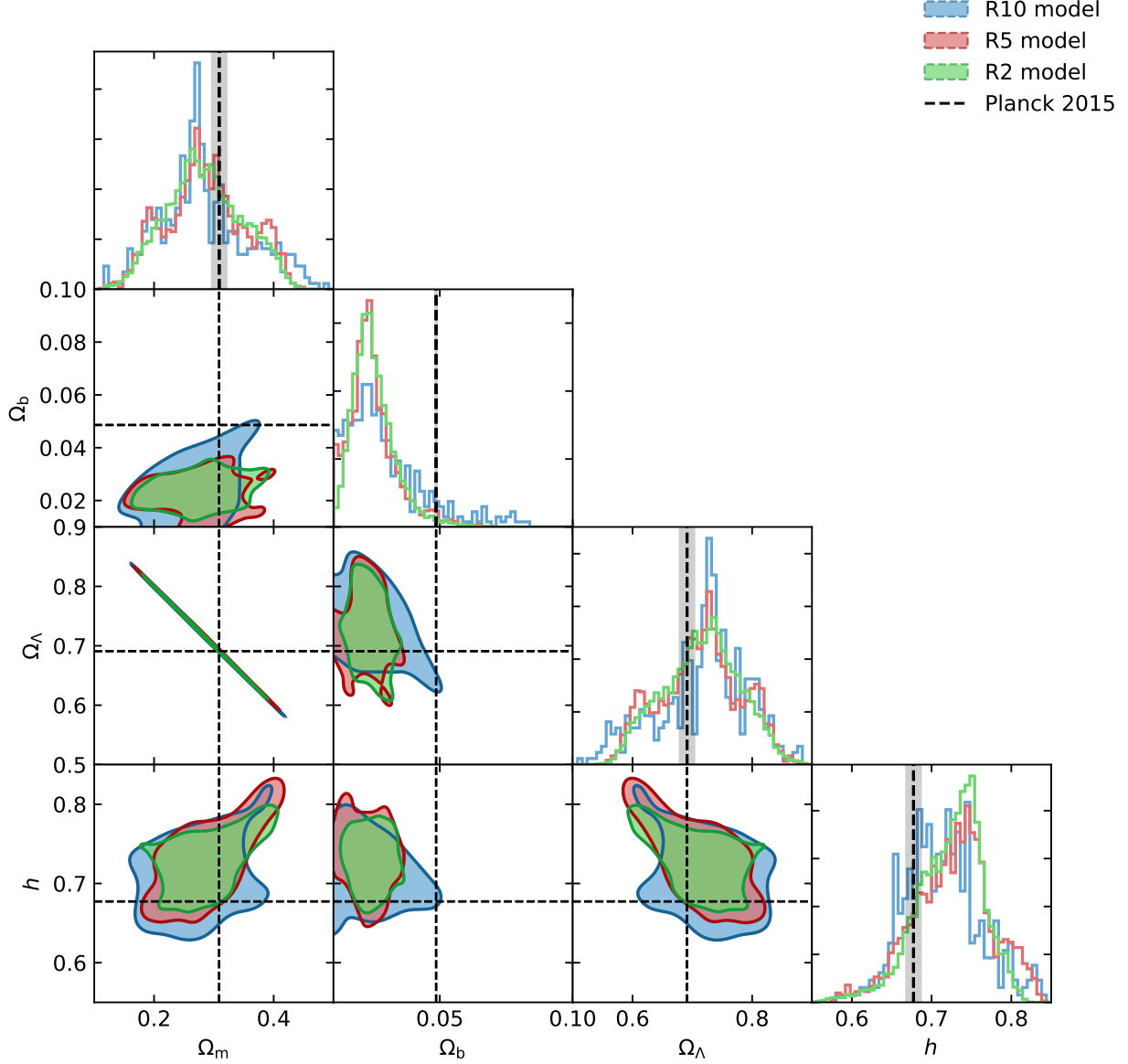


Figure 27: Inference results of the R2, R5 and R10 models the observed SDSS spectra. The predictions for Ω_m and Ω_Λ are in agreement with the Planck measurements, while h also is in agreement with the Planck value, although our models seem to favor higher values. Ω_b is significantly underestimated by our models.

The predicted values of Ω_m and Ω_Λ are consistent with the measurements reported by the Planck 2015 analysis, $\Omega_{m,\text{Planck}} = (0.309 \pm 0.012)$ and $\Omega_{\Lambda,\text{Planck}} = (0.691 \pm 0.012)$, within the 1σ confidence interval. Nevertheless, the inferred distributions show a slight tendency towards lower values of Ω_m and correspondingly higher values of Ω_Λ .

For the reduced Hubble parameter, our estimate $h = (0.721 \pm 0.026)$ is consistent with the Planck value $h_{\text{Planck}} = (0.677 \pm 0.009)$ within the combined uncertainties. However, the peaks of the predicted distributions seem to favor a somewhat larger value of h . Interestingly, measurements based on local distance ladder observations, such as those reported by the SH0ES collaboration, obtain $h_{\text{SH0ES}} = (0.7253 \pm 0.0099)$, which lies closer to the value favored by our pipeline.

This difference reflects the well-known Hubble tension, referring to the discrepancy between measurements of the Hubble parameter inferred from early-universe observations such as the cosmic microwave background (CMB) and those obtained from late-universe probes like distance ladder measurements. Given the relatively large uncertainties of our measurement, it is not possible to draw strong conclusions about this tension from our results alone. Nevertheless, the fact that Lyman- α forest spectra contain information about h highlights their potential as an independent probe of cosmological parameters.

The inference of the baryon density shows a noticeable bias toward lower values, significantly underestimating the Planck measurement of $\Omega_{b,\text{Planck}} = (0.0486 \pm 0.0002)$. This discrepancy is visible in Fig. 27, where the predicted distribution peaks well below the Planck value. This hints to a discrepancy between our modeling of Lyman- α spectra and reality, as inference on the reference box did not show this behavior. As Ω_b does not have a strong influence on the evolution of the universe, we expect the model to mainly learn properties like the mean flux to infer this parameter. In fact, we do find a systematic offset when comparing the mean transmitted flux of the simulated spectra with that of the SDSS data. We will return to this point and possible explanations for this behavior, such as the choice of UVB in the simulations, in the discussion.

Overall, these results demonstrate that the developed pipeline can successfully extract cosmological information from observed Lyman- α forest spectra, while also showing systematic deviations that will be examined further in the following section.

5 Discussion

In this work, we developed a simulation-based inference pipeline using a Transformer neural network to predict cosmological parameters directly from Lyman- α forest spectra. The model was trained on a large set of simulated spectra, allowing it to learn the relationship between spectral absorption features and the underlying cosmological parameters without relying on summary statistics such as the flux power spectrum. Applying this pipeline to both simulated test datasets and observed SDSS spectra, we demonstrated that the model is capable of retrieving several cosmological parameters with reasonable accuracy. In particular, the inferred values of Ω_m and Ω_Λ are consistent with constraints from the Planck 2015 survey. At the same time, the analysis reveals systematic deviations in the inference of Ω_b and a weak preference for higher values of the dimensionless Hubble parameter h .

To investigate the bias in Ω_b , we examine the distributions of the mean transmitted flux for individual spectra in both the training datasets and the observed SDSS spectra (Fig. 28). As the baryon density primarily determines the amount of absorbing gas in the intergalactic medium, the network is expected to rely on global spectral properties such as the mean flux when inferring Ω_b . The comparison reveals a clear discrepancy between the simulated and observed spectra, indicating that the simulations systematically predict different mean flux levels than those measured in the SDSS data. A likely origin of this difference is the assumed ultraviolet background (UVB) model used in the simulations. Because radiative transfer is not modeled directly, the UVB determines the photoionization rate of hydrogen and therefore controls the neutral hydrogen fraction in the intergalactic medium. Consequently, the UVB strongly affects the mean transmitted flux of the spectra. The sensitivity of Lyman- α forest observables to the UVB is a well-known issue in Lyman- α modeling and highlights the importance of accurately calibrating the UVB when comparing simulations to observations.

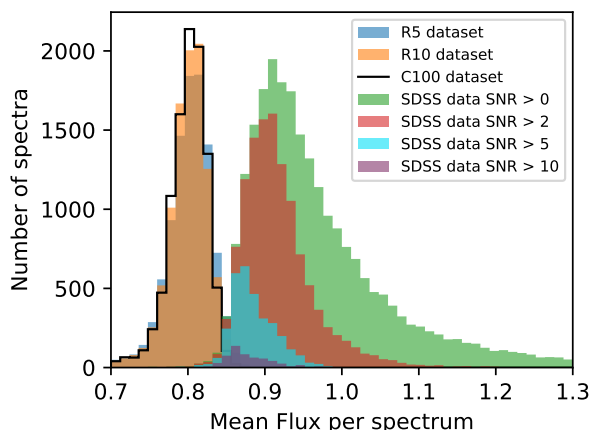


Figure 28: Comparison of mean flux per spectrum between the simulated spectra (R5, R10 and C100 datasets) and the observed SDSS spectra. There is a clear discrepancy between them, likely caused by the assumed UVB model in the simulations.

The inferred distributions for h are generally broad, indicating that the parameter is only weakly constrained within our current setup. Nevertheless, the predicted distributions tend to peak at somewhat higher values of h , which are closer to those reported by the SH0ES collaboration using local distance ladder measurements than to the value inferred from Planck CMB observations. In the context of the Hubble tension, this behavior could in principle be interesting. However, the uncertainties of our measurement are large, and the simulations used in this work cover only relatively small cosmological volumes. As a result, the spectra contain limited information about the large-scale expansion history of the universe, which is the primary driver of constraints on h . It is therefore possible that the network partly relies on parameter correlations learned during training

rather than directly extracting independent information about h from the spectra. For these reasons, our analysis does not allow any meaningful conclusions regarding the Hubble tension. Nonetheless, the observed tendency toward higher values of h may be an interesting direction to explore in future studies with larger simulation volumes.

There are several directions future works could pursue to improve the pipeline developed in this study. One important direction would be to increase the physical realism of the simulated spectra used during training, in particular through improved modeling of the ultraviolet background or calibrating the simulated spectra to match the observed mean flux levels. This could help reduce systematic biases such as the one observed for Ω_b . Another promising extension would be to increase the simulation volume and include snapshots from multiple redshifts in the spectra generation. This would provide the model with access to larger-scale cosmological information and a broader range of physical conditions in the intergalactic medium, which could improve constraints on parameters such as the dimensionless Hubble parameter h . More generally, the results presented here demonstrate that the Transformer-based approach can successfully extract cosmological information from Lyman- α forest spectra, and future work building on this framework may further enhance the precision and robustness of such analyses.

6 Appendix

6.1 Universe Transitions

Given the Friedmann Equation

$$H^2 = H_0^2(\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda), \quad (11)$$

where $H = \dot{a}/a$, it is immediately clear that the expansion of the universe is dominated by different components for different intervals of the scale factor $a(t)$. Unfortunately, the solution to the Friedmann Equation can not be expressed by elementary functions, and thus it is well motivated and a good approximation to calculate the transition points for single component universes. For a radiation dominated and matter dominated universe, the Friedmann Equation reduces to

$$H^2 = H_0^2 \Omega_r a^{-4} \quad (12)$$

$$H^2 = H_0^2 \Omega_m a^{-3} \quad (13)$$

respectively. We now want to find the transition point, when both components contribute to the expansion equally, and so we obtain

$$H_0^2 \Omega_r a^{-4} = H_0^2 \Omega_m a^{-3} \quad (14)$$

and thus

$$a = \frac{\Omega_r}{\Omega_m} \approx 1.84 \times 10^{-4} \quad (15)$$

$$\Rightarrow z \approx 3700. \quad (16)$$

6.2 The SBI Pipeline in a Measure Theory Setting

Given the measurable space of all parameters (Θ, Σ_θ) and the measurable space of all data $(\mathcal{X}, \Sigma_\mathcal{X})$, we can express the probability distribution $p(x \mid \theta_*)$ as the probability measure

$$\mu_x^{\theta_*} = p(x \mid \theta_*), \quad (17)$$

where $\mu_x^{\theta_*} : \Sigma_\mathcal{X} \rightarrow [0, 1]$. This measure can be applied to an element of the σ -algebra $A \in \Sigma_\mathcal{X}$ by calculating

$$\mu_x^{\theta_*} = \int_A p(x \mid \theta_*) dx. \quad (18)$$

Our Transformer network now learns a measurable map $f : \mathcal{X} \rightarrow \Theta$, which can be used to define the push forward measure of $\mu_x^{\theta_*}$ in the following sense

$$f_{\#} \mu_x^{\theta_*}(B) = \int_{f^{-1}(B)} p(x \mid \theta_*) dx, \quad (19)$$

where $B \in \Sigma_\Theta$ and $f_{\#} \mu_x^{\theta_*} : \Sigma_\Theta \rightarrow [0, 1]$. In this framework, training the model can now be thought of as selecting a measurable map f such that, for each data-generating measure $\mu_x^{\theta_*}$, the induced estimator measure $f_{\#} \mu_x^{\theta_*}$ minimizes expected squared distance² to the true parameter θ_* on the space Θ .

²The squared distance comes from the loss function. More generally, training minimizes the loss function.

6.3 Spectral Width Derivation

To derive an equation for the width of the spectra, we first need the relation between redshift z and scale factor a . To do so, we assume a matter and Λ dominated universe, as curvature is set to zero and the energy density of radiation can be approximated to be zero (see Appendix 6.1). This gives us the Friedmann equation of the form

$$\dot{a} = H_0^2(\Omega_m a^{-1} + \Omega_\Lambda a^2), \quad (20)$$

which is a non linear first order ODE, that can be solved analytically. To do so, we perform a separation of variables, giving us

$$H_0 t = \int \frac{1}{\sqrt{\frac{\Omega_m}{a} + \Omega_\Lambda a^2}} da = \int \frac{\sqrt{a}}{\sqrt{\Omega_m + \Omega_\Lambda a^3}} da \quad (21)$$

which we can bring into the form

$$H_0 t = \frac{2}{3\sqrt{\Omega_\Lambda}} \int \frac{1}{\sqrt{1+u^2}} du, \quad (22)$$

by substituting $u = \sqrt{\frac{\Omega_\Lambda a^3}{\Omega_m}}$. The integral expression is a standard integral, which can be expressed as

$$\int \frac{1}{\sqrt{1+u^2}} du = \ln(\sqrt{u^2+1} + u) + C = \text{arsinh}(u) + C. \quad (23)$$

Given the initial conditions of $a(t=0) = 0$, we find $C = 0$. Thus we get the following two expressions from 22:

$$t(a) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \text{arsinh} \left(\sqrt{\frac{\Omega_\Lambda a^3}{\Omega_m}} \right) \quad (24)$$

$$a(t) = \sqrt[3]{\frac{\Omega_m}{\Omega_\Lambda} \sinh^2 \left(\frac{3H_0\sqrt{\Omega_\Lambda}t}{2} \right)}. \quad (25)$$

To now calculate the width of the spectra, we first need to understand, that the width of the spectra is purely determined by the amount of redshift the Lyman- α wavelength experiences in the time t_{tr} it takes light to travel through the box. We thus first need to calculate the time it takes light to travel through the box once:

$$t_{tr} = \frac{d_p}{c} = d_c \frac{a(t)}{c}, \quad (26)$$

where c is the speed of light, d_p is the physical size of the box, and d_c is the box size in comoving coordinates. In this equation, we assume the box to have a fixed size during the time the light travels through the box. This is true for the simulation snapshots and thus the creation of spectra with TEMET, however is not true physically and thus an approximation that introduces a small ($\approx 1\%$) systematic error.

With Eq. 25 and Eq. 26 at hand, we can finally calculate the difference in redshift between the time t_{start} the light started its journey on one side of the box and the time $t_{start} + t_{tr}$ it reached the other side:

$$\Delta z = z(t_{start}) - z(t_{start} + t_{tr}) = z(t_{start}) - \left(\frac{1}{a(t_{start} + t_{tr})} - 1 \right). \quad (27)$$

Plugging in Eq. 25 and writing $z(t_{start})$ simply as z , we get

$$\Delta z = z + 1 - \frac{1}{\sqrt[3]{\frac{\Omega_m}{\Omega_\Lambda} \sinh^2 \left(\frac{3}{2} H_0 \sqrt{\Omega_\Lambda} (t_{start} + t_{tr}) \right)}}. \quad (28)$$

We can further plug Eq. 24 in for t_{start} and Eq. 26 for t_{tr} :

$$\Delta z = z + 1 - \frac{1}{\sqrt[3]{\frac{\Omega_m}{\Omega_\Lambda} \sinh^2 \left(\frac{3}{2} H_0 \sqrt{\Omega_\Lambda} \left(\frac{2}{3H_0\sqrt{\Omega_\Lambda}} \text{arsinh} \left(\sqrt{\frac{\Omega_\Lambda a(t_{start})^3}{\Omega_m}} \right) + d_c \frac{a(t_{start})}{c} \right) \right)}}}, \quad (29)$$

and then use $a(t_{start}) = 1/(1+z)$ to obtain

$$\Delta z = z + 1 - \frac{1}{\sqrt[3]{\frac{\Omega_m}{\Omega_\Lambda} \sinh^2 \left(\frac{3}{2} H_0 \sqrt{\Omega_\Lambda} \left(\frac{2}{3 H_0 \sqrt{\Omega_\Lambda}} \operatorname{arsinh} \left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m (1+z)^3}} \right) + d_c \frac{1}{c(1+z)} \right) \right)}}. \quad (30)$$

Simplifying this expression we finally arrive at

$$\Delta z = z + 1 - \frac{1}{\left(\frac{\Omega_m}{\Omega_\Lambda} \right)^{1/3} \sinh^{2/3} \left(\frac{3 H_0 \sqrt{\Omega_\Lambda} d_c}{2 c (1+z)} - \operatorname{arsinh} \left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m (1+z)^3}} \right) \right)}. \quad (31)$$

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Declaration

In this work, the large language model-based chatbot ChatGPT was used to improve language clarity and structure and to suggest code snippets for data visualization. I understand that the use of products generated by AI-based tools does not guarantee their quality and have reviewed, adapted and verified all generated suggestions. All scientific analyses, simulations, figures, and interpretations were carried out by me, and I bear full responsibility for the accuracy of the content presented in this thesis.

I hereby declare that I have written this thesis independently and have not used any sources or aids other than those specified.

Heidelberg, March 15, 2026

A handwritten signature in black ink, appearing to read 'J. Boucsein', written below a horizontal line.

Jeremias Nils Boucsein